



ALLAHABAD UNIVERSITY

NOTES

FOR

BSC 3<sup>RD</sup> YEAR

ZOLOGY

PYQ BASED



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year questions



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## Question/Answer

**Question:-** Separation of the two strands of the DNA helix on heating is called?

**Answer:** Denaturation (or DNA melting).

**Question:** Based on renaturation kinetics, the DNA can be classified into ....., ..... and ..... sequences.

**Answer:** Highly repetitive, moderately repetitive, and unique (single-copy) sequences.

**Question:** DNA sequences which have the unique ability to move as discrete units from one position in the genome to another are called?

**Answer:** Transposons (or transposable elements / jumping genes).

**Question:** Method of hybridizing DNA to nitrocellulose bound single-stranded DNA is known as?

**Answer:** Southern blotting.

**Question:** The enzyme required for the synthesis of cDNA from mRNA is?

**Answer:** Reverse transcriptase (RNA-dependent DNA polymerase).

**Question:** ..... and ..... studied the *E. coli* lactose system and gave the ..... for regulation of gene expression in prokaryotes.

**Answer:** François Jacob, Jacques Monod, and Operon model.

**Question:** ..... found as extrachromosomal circular DNA molecules in most bacteria can be used as ..... in Recombinant DNA technology.

**Answer:** Plasmids and cloning vectors.

**Question:** Sequences recognized by many restriction enzymes are ..... which show complementary bases on the two sides of .....

**Answer:** Palindromic sequences and axis of symmetry.

**Question:** ..... is currently being used in forensic science for identification of crime suspects.

**Answer:** DNA Profiling (or DNA Fingerprinting).

**Question:** ..... are proteins of the immune system and their function is to interact with specific foreign molecules called ..... and render them inactive.

**Answer:** Antibodies (Immunoglobulins) and antigens.

**Question:** When Thymine is substituted for adenine it is known as?

**Answer:** Transversion mutation.

**Question:** Genetic recombination in bacteria mediated by bacteriophage is known as?

**Answer:** Transduction.

**Question:** ..... is a segment of DNA in which the base-pair sequence reads the same in both directions from a point of symmetry.

**Answer:** Palindromic DNA.

**Question:** ..... are intervening sequences of DNA within eukaryotic gene and are not present in mature mRNA.

**Answer:** Introns.

**Question:** The correct structure of DNA was first deduced by ..... and .....

**Answer:** James Watson and Francis Crick.

**Question:** Antibodies belong to the class of proteins called?

**Answer:** Immunoglobulins (gamma-globulins).

**Question:** A bacterium harbouring a prophage is said to be?

**Answer:** Lysogenic (or a lysogen).

**Question:** A nucleus which is formed by the fusion of nuclei is called a?

**Answer:** Syngamy (or zygotic nucleus).

**Question:** *Malignant tertian malaria* is caused by?

**Answer:** *Plasmodium falciparum*.

**Question:** *Monocystis* is found in ..... of earthworm.

**Answer:** Seminal vesicles.

**Question:** *Callosobruchus chinensis* is a pest of?

**Answer:** Stored pulses (pulse beetle).

**Question:** Causative organism of tick fever is?

**Answer:** *Babesia bigemina*.

**Question:** Which honey bee species produces maximum honey per comb?

**Answer:** *Apis dorsata* (rock honey bee).

**Question:** Pebrine is a disease of?

**Answer:** Silkworms (*Bombyx mori*).

**Question:** *Trichogramma* is a .....?

**Answer:** Egg parasitoid.

**Question:** ICUN stands for?

**Answer:** International Union for Conservation of Nature (officially IUCN).

**Question:** Scabies is caused by?

**Answer:** *Sarcoptes scabiei* (itch mite).

**Question:** Celphos is used for the control of?

**Answer:** Stored grain pests and rodents.

**Question:** A nucleotide can be described as a ..... nucleoside.

**Answer:** Phosphorylated nucleoside.

**Question:** Removal of introns in eukaryotic mRNA processing takes place by the formation of?

**Answer:** Spliceosomes (lariat structure).

**Question:** Transduction is mediated by?

**Answer:** Bacteriophages.

**Question:** Phages involved in lysogenic relationships are called?

**Answer:** Temperate phages.

**Question:** Antibody molecules are also called?

**Answer:** Immunoglobulins (Ig).

**Question:** *Plasmodium falciparum* causes ..... malaria

**Answer:** Malignant tertian malaria.

**Question:** ..... mosquito is the vector of encephalitis.

**Answer:** *Culex* mosquito.

**Question:** Bioagents used in biological control of pests are ....., parasitoids, and predators.

**Answer:** Pathogens.

**Question:** The process of soil formation is termed as?

**Answer:** Pedogenesis.

**Question:** The presence of unwanted sound at a place is known as?

**Answer:** Noise pollution.

**Question:** The father of Chipko movement is?

**Answer:** Chandi Prasad Bhatt (and Sunderlal Bahuguna).

**Question:** A few years back, the most common heavy metal at the places of heavy traffic used to be? **Answer:** Lead ( $\text{Pb}$ ).

**Question:** The study of impact of xenobiotics at ecosystem level is known as?

**Answer:** Ecotoxicology.

**Question:** Conservation of wildlife in their natural habitat is ..... conservation.

**Answer:** *In-situ* conservation.

**Question:** Cultivation of aquatic animals for food and products is known as?

**Answer:** Aquaculture.

**Question:** DNA polymerase has ..... sites for attachment.

**Answer:** Multiple functional domain sites.

**Question:** Denaturation of DNA means separation of two strands of DNA by .....

**Answer:** Disrupting hydrogen bonds (via heat or chemicals).

**Question:** Circular double-stranded DNA molecules found in bacterial cells are called?

**Answer:** Plasmids.

**Question:** There are ..... distinct types of restriction endonucleases.

**Answer:** Four distinct types (Type I, II, III, IV).

**Question:** Interleukin-4 is produced by ..... cells.

**Answer:** T-helper 2 ( $\text{Th}_2$ ) cells.

**Question:** In bacteria, the gene which produces a repressor substance that binds to the operator gene, is called ..... gene.

**Answer:** Regulator gene (or i-gene).

**Question:** The synthesis of new RNA molecules using DNA as a template is called?

**Answer:** Transcription.

**Question:** ..... is a process of virus mediated transfer of genetic material from one host cell to another.

**Answer:** Transduction.

**Question:** cDNA is produced usually by using ..... as a template.

**Answer:** Messenger RNA (mRNA).

**Question:** The polymerase chain reaction (PCR) technique is used for the gene .....

**Answer:** Amplification.

### Question: Define Recombinant DNA.

#### Answer:

- **Definition:** Recombinant DNA (rDNA) is an artificial DNA molecule created in a laboratory by combining genetic material from two or more different organisms.
- **Mechanism:** Target DNA fragments are isolated using **restriction endonucleases** and linked to a vector using **DNA ligase**.
- **Significance:** It serves as the foundation for genetic engineering, enabling foreign gene expression in host organisms for applications like insulin production and gene therapy.

### Question: Define Cosmid.

#### Answer:

- **Definition:** A cosmid is a hybrid cloning vector constructed by integrating plasmid features with the **cos sites** (cohesive end sites) derived from the **lambda phage**.
- **Cloning Capacity:** It can accommodate large inserts of foreign DNA ranging from **35 to 45 kb**.
- **Function:** Cosmids replicate autonomously inside bacterial hosts like *E. coli* as plasmids or can be packaged *in vitro* into phage particles for efficient transfection.

### Question: Define Humoral Immunity.

#### Answer:

- **Definition:** Humoral immunity is the component of the adaptive immune response mediated by macromolecules, primarily **antibodies (immunoglobulins)** circulating in extracellular fluids such as blood plasma and lymph.
- **Cellular Origin:** It is driven by **B-lymphocytes**, which differentiate into antibody-secreting **plasma cells** upon antigen recognition.
- **Function:** Antibodies recognize, bind to, and neutralize extracellular pathogens (bacteria, viruses) and soluble toxins.

### Question: What is Repression in Gene Regulation?

#### Answer:

- **Definition:** Repression is a regulatory mechanism of gene expression where the transcription of a gene or operon is inhibited by a specific regulator protein called a **repressor**.
- **Mechanism:** The repressor protein binds to a specific **operator sequence** near the promoter, physically blocking **RNA polymerase** from transcribing the target genes.
- **Example:** In the *E. coli* **lac operon**, the Lac repressor inhibits transcription in the absence of lactose.

### Question: Define Transposons.

#### Answer:

- **Definition:** Transposons (or **transposable elements** / "**jumping genes**") are mobile

DNA sequences capable of moving or copying themselves to different locations within a genome.

- **Mechanism:** Their movement is catalyzed by the enzyme **transposase** and is typically flanked by inverted repeat sequences.
- **Significance:** Discovered by Barbara McClintock, transposons cause genome rearrangements and mutations, driving genetic diversity.

## Question: What are Split Genes?

Answer:

- **Definition:** Split genes are eukaryotic genes whose coding sequences are interrupted by non-coding intervening sequences.
- **Key Components:**
  - **Exons:** Coding sequences that are retained in mature mRNA and expressed as proteins.
  - **Introns:** Non-coding regions that are transcribed into primary transcripts (hnRNA) but removed via **RNA splicing**.

## Question: What is Gene Cloning?

Answer:

- **Definition:** Gene cloning is the molecular technique of making multiple identical copies of a specific gene or DNA sequence using recombinant DNA technology.
- **Key Steps:**
  1. Isolation of target DNA fragment.
  2. Ligation of target DNA into a vector plasmid.
  3. Transformation of the recombinant vector into a host organism (e.g., *E. coli*).
  4. Selection and multiplication of transformed host cells.

## Fill in the Blanks

Question: Fill in the blanks with appropriate terms:

1. In eukaryotes, the sequences found between the coding sequences of a gene are known as \_\_\_\_\_.
  - Answer: Introns
2. Myasthenia gravis is an \_\_\_\_\_ disease.
  - Answer: Autoimmune
3. Hybridomas are screened using \_\_\_\_\_ media.
  - Answer: HAT (Hypoxanthine-Aminopterin-Thymidine)
4. Recombinant insulin can be obtained from a bacterial cell, provided the insulin gene is recombined to an \_\_\_\_\_ vector.
  - Answer: Expression
5. Gilbert method can be used for \_\_\_\_\_ sequencing.
  - Answer: DNA
6. Hybridization between DNA and DNA is known as \_\_\_\_\_.
  - Answer: Southern Hybridization / Southern Blotting

7. The part of a gene which gets expressed is known as \_\_\_\_\_.
  - Answer: Exon
8. Gene amplification is done by \_\_\_\_\_ technique.
  - Answer: PCR (Polymerase Chain Reaction)
9. \_\_\_\_\_ is used to remove 5'-\text{PO}\_4 in a DNA molecule.
  - Answer: Alkaline Phosphatase
10. The part of a bacterium that contains nucleic acid and is analogous in function to the nucleus of a eukaryotic cell is known as \_\_\_\_\_.
  - Answer: Nucleoid
11. \_\_\_\_\_ enzymes can recognize specific DNA sequences.
  - Answer: Restriction Endonuclease
12. \_\_\_\_\_ is a segment of DNA in which the base pair sequence reads the same in both directions from a point of symmetry.
  - Answer: Palindrome / Palindromic Sequence
13. A cell surface antigen, which was first discovered in rhesus monkey is called \_\_\_\_\_ blood group.
  - Answer: Rh (Rhesus)
14. Asthma is a common manifestation of \_\_\_\_\_.
  - Answer: Allergy / Type-I Hypersensitivity
15. Transduction is mediated by \_\_\_\_\_.
  - Answer: Bacteriophages (Bacterial Viruses)

**Question: Expand and define the following acronyms:**

- **hnRNA:**
  - **Full Form:** Heterogeneous nuclear RNA
  - **Definition:** Precursor mRNA found in the eukaryotic nucleus prior to post-transcriptional processing (splicing, capping, and polyadenylation).
- **HAT:**
  - **Full Form:** Hypoxanthine-Aminopterin-Thymidine
  - **Definition:** A selective cell culture medium used in hybridoma selection during monoclonal antibody production.
- **dNTP:**
  - **Full Form:** Deoxyribonucleoside Triphosphate
  - **Definition:** Substrate building blocks (dATP, dCTP, dGTP, dTTP) utilized by **DNA polymerase** for DNA synthesis.
- **DNA pol:**
  - **Full Form:** DNA Polymerase
  - **Definition:** The enzyme responsible for synthesizing new complementary DNA strands during replication and repair.
- **snRNA:**
  - **Full Form:** Small nuclear RNA
  - **Definition:** Non-coding nuclear RNA molecules that combine with proteins to form **spliceosomes** for pre-mRNA splicing.

### Question: Write a short note on Restriction Mapping.

#### Answer:

- **Definition:** Restriction mapping is a method used to map the relative positions of restriction enzyme cleavage sites on a specific DNA molecule.
- **Methodology:** DNA is digested separately and in combination with specific **restriction enzymes**, and the fragment lengths are separated using **agarose gel electrophoresis**.
- **Application:** Used for gene cloning, characterization of plasmids, and verifying recombinant constructs.

### Question: Write a short note on Repetitive DNA Sequences.

#### Answer:

- **Definition:** Repetitive DNA sequences are nucleotide sequences present in multiple copies throughout a genome.
- **Classifications:**
  - **Tandem Repeats:** Satellite DNA, Minisatellites (VNTRs), and Microsatellites (STRs).
  - **Interspersed Repeats:** Dispersed elements like SINEs, LINEs, and Transposons.
- **Importance:** Plays a structural role in heterochromatin and is essential for **DNA fingerprinting** and linkage mapping.

### Q) Expand the acronym: hnRNA

**Answer:Full Form:** Heterogeneous nuclear RNA.

- **Explanation:** Primary, unprocessed pre-mRNA transcript synthesized inside the eukaryotic nucleus by RNA Polymerase II. It contains both coding sequences (**exons**) and non-coding sequences (**introns**) prior to post-transcriptional modifications.

### Q)Expand the acronym: HAT

**Answer:Full Form:** Hypoxanthine-Aminopterin-Thymidine Medium.

- **Explanation:** Selective cell culture medium utilized in hybridoma technology to isolate fused hybrid cells (B-lymphocyte + myeloma cell). Aminopterin blocks *de novo* nucleotide synthesis, forcing viable cells to utilize the salvage pathway via hypoxanthine and thymidine.

### Q)Expand the acronym: dNTP

**Answer:Full Form:** Deoxyribonucleoside Triphosphate.

- **Explanation:** High-energy nucleoside precursors ( $\text{dATP, dCTP, dGTP, dTTP}$ ) acting as essential substrate building blocks for DNA polymerase during DNA replication and PCR amplification.

### Q)Expand the acronym: MHC

**Answer:Full Form:** Major Histocompatibility Complex.

- **Explanation:** Cell-surface glycoprotein complex responsible for binding pathogen-derived peptide fragments and displaying them to T-lymphocytes for adaptive antigen recognition.

### Q)Expand the acronym: snRNA

**Answer:Full Form:** Small nuclear RNA.

- **Explanation:** Non-coding eukaryotic nuclear RNA molecules that complex with specific proteins to form **snRNPs** (small nuclear ribonucleoproteins), which assemble the spliceosome complex to excise introns during RNA processing.

### Q)Define: Recombinant DNA

**Answer:Definition:** A chimeric DNA strand constructed *in vitro* by joining genetic sequences derived from two or more distinct biological species or sources.

- **Key Mechanism:** Produced by cleaving foreign target DNA and vector DNA with restriction endonucleases and joining the fragments via DNA ligase.

### Q)Define: Cosmid

**Answer:Definition:** A hybrid cloning vector constructed by incorporating the *cos* (cohesive end) sequence of Bacteriophage lambda ( $\lambda$ ) into a bacterial plasmid vector.

### Q) Define: Humoral Immunity

**Answer:Definition:** The arm of adaptive immunity mediated by circulating extracellular antibodies (immunoglobulins) present in bodily fluids like plasma and lymph.

- **Effector Cells:** Driven by differentiated B-lymphocytes (plasma cells) that neutralize toxins and targeted extracellular pathogens.

### Q). Define: Repressor

**Answer:Definition:** A regulatory DNA-binding protein synthesized by a regulator gene that halts gene transcription.

- **Mechanism:** Binds specifically to an operator locus to sterically hinder RNA polymerase from binding or moving along structural genes (e.g., *Lac* repressor).

### Q) Define: Transposons

**Answer:Definition:** Discrete mobile DNA sequences ("jumping genes") capable of moving to new positions within a host genome.

- **Significance:** Catalyzed by the enzyme **transposase**, transposons induce genomic mutations, chromosomal rearrangements, and genetic variation.

### Q) Which of the following is NOT used as a vector in Recombinant DNA Technology?

(a) Plasmids    (b) Cosmids    (c) Bacteriophage    (d) Bacteria

**Answer:Correct Option: (d) Bacteria**

- **Reason:** Bacteria serve as **host organisms** that receive foreign DNA, whereas plasmids, cosmids, and bacteriophages function as gene transfer vehicles (vectors).

### Q) A DNA molecule formed by the insertion of foreign DNA is called:

(a) Chimaeric DNA    (b) Unique DNA    (c) Repetitive DNA    (d) Satellite DNA

**Answer:Correct Option: (a) Chimaeric DNA**

- **Reason:** A chimeric (or recombinant) strand contains combined genetic sequences originating from different organisms.

### Q)Cot curve is associated with:

(a) Replication of DNA    (b) Repair of DNA    (c) Reassociation kinetics of DNA    (d) Damage of DNA

**Answer:Correct Option: (c) Reassociation kinetics of DNA**

- **Reason:** A  $C_{0t}$  curve measures the rate at which denatured single-stranded DNA re-anneals, evaluating total genome complexity and repetition level.

### Q) B cells and T cells form the cellular components of:

(a) Digestive system    (b) Immune system    (c) Respiratory system    (d) Excretory system

**Answer:**

- **Correct Option: (b) Immune system**
- **Reason:** B cells (humoral response) and T cells (cell-mediated response) are specialized lymphocytes mediating adaptive immunity.

**Q)Poly (A) tail formation in the mRNA occurs:**

(a) At the 5' end    (b) At the 3' end    (c) In the intron    (d) In the exon

**Answer:**

- **Correct Option: (b) At the 3' end**
- **Reason:** Polyadenylation adds 200\text{--}300 adenine residues to the 3'-OH terminus of eukaryotic pre-mRNA to prevent enzymatic degradation and enable nuclear export.

**Q)The two strands of DNA are held together by:**

(a) Phosphodiester bond    (b) Phosphoanhydride bond    (c) Hydrogen bonds    (d) C-C covalent bonds

**Answer:**

- **Correct Option: (c) Hydrogen bonds**
- **Reason:** Complementary nitrogenous base pairs across opposing strands are joined by hydrogen bonds (2 between Adenine–Thymine, 3 between Guanine–Cytosine).

**Q)Which one of the following best represents a gene clone?**

(a) Multiple copies of a gene kept in a suitable medium    (b) Recombinant gene kept in a suitable medium    (c) Clone of bacterial cell inserted with a gene    (d) Clone of a bacterial cell carrying a recombined gene

**Answer:**

- **Correct Option: (a) Multiple copies of a gene kept in a suitable medium**
- **Reason:** A gene clone represents an identical population of exact copies derived from a single ancestral target gene.

**Q)Degeneracy of genetic code is explained by:**

(a) Wobble hypothesis    (b) Signal hypothesis    (c) Chargaff rule    (d) C-value paradox

**Answer:Correct Option: (a) Wobble hypothesis**

- **Reason:** Proposed by Francis Crick, it shows that non-standard base pairing at the 3rd codon position allows a single tRNA anticodon to recognize multiple synonymous codons.

**Q) DNA segments formed by reverse transcription are called .....**

**Answer:Term: cDNA (Complementary DNA)**

- **Explanation:** Synthesized *in vitro* from a mature single-stranded mRNA template through reverse transcriptase activity.

**Q) Hybridization of DNA sequences with RNA is done by .....**

**Answer:Term: Northern Blotting**

- **Explanation:** Involves transferring gel-separated RNA fragments onto a membrane, followed by hybridization with labeled complementary single-stranded DNA probes.

**Q)Transcription in prokaryotes is coupled with .....**

**Answer:Term: Translation**

- **Explanation:** Because prokaryotes lack a nuclear membrane barrier, ribosomes attach to nascent mRNA to begin protein synthesis while transcription is actively occurring.

**Q) Plasmids are used as ..... in Recombinant DNA technology.**

**Answer:Term: Cloning Vectors**

- **Explanation:** Extra-chromosomal, circular, self-replicating DNA molecules act as vehicles to transfer target genes into host microorganisms.

**Q)..... are cellular components of the immune system.**

**Answer:Term: Lymphocytes (B cells and T cells)**

- **Explanation:** Principal adaptive white blood cells that mediate antibody secretion and cell-mediated pathogen destruction.

**Q)..... is a disease of the immune system.**

**Answer:Term: AIDS (Acquired Immunodeficiency Syndrome)**

- **Explanation:** A chronic disease caused by the Human Immunodeficiency Virus (HIV), which selectively destroys CD4+ T-helper cells.

**Q)..... are formed during the removal of introns in mRNA processing.**

**Answer:Term: Lariat structures**

- **Explanation:** A loop-shaped excised RNA intermediate produced during spliceosome-mediated nuclear pre-mRNA splicing.

**Q)DNA sequences which can change places in the genome are called .....**

**Answer:Term: Transposons (or Transposable Elements)**

- **Explanation:** Mobile genomic sequences capable of self-directed excision and re-insertion into alternate chromosomal loci.

**Q)..... is used for the production of monoclonal antibodies.**

**Answer:Term: Hybridoma Technology**

- **Explanation:** Fuses specific antibody-producing plasma B-cells with immortal myeloma cells to create continuous monoclonal cell lineages.

**Q) DNA sequences which are mirror images of each other from the axis of symmetry are called .....**

**Answer:Term: Palindromic sequences**

- **Explanation:** Symmetrical sequence regions reading identically on complementary strands when parsed in the 5'  $\rightarrow$  3' direction.

**Q) DNA polymerase has ..... sites for attachment.**

**Answer:Term: Multiple functional domains**

- **Explanation:** Contains specialized attachment domain sites including the template-primer binding pocket, catalytic polymerase site, and 3'  $\rightarrow$  5' exonuclease domain.

**Q) Denaturation of DNA means separation of two strands of DNA by .....**

**Answer:Term: Disrupting Hydrogen Bonds**

- **Explanation:** Physical heat or alkaline chemicals break hydrogen bonds linking nitrogenous bases without cleaving covalent phosphodiester backbones.

**Q) Circular double-stranded DNA molecules found in bacterial cells are called .....**

**Answer:Term: Plasmids**

- **Explanation:** Autonomously replicating extrachromosomal circular DNA molecules carrying non-essential beneficial traits (e.g., antibiotic resistance).

**Q)There are ..... distinct types of restriction endonucleases.**

**Answer:Term: Four (Type I, II, III, IV)**

- **Explanation:** Categorized according to subunit structure, cleavage position relative to recognition sites, and specific cofactor requirements.

**Q)Interleukin-4 is produced by ..... cells.**

**Answer:Term: Th2 (Type-2 Helper T)**

- **Explanation:** An immunomodulatory cytokine regulating B-cell proliferation, differentiation, and IgE antibody class switching.

**Q) In bacteria, the gene which produces a repressor substance that binds to the operator gene, is called ..... gene.**

**Answer:Term: Regulator (i) gene**

- **Explanation:** Continuously transcribes repressor proteins to keep the operon in an "off" state in the absence of an inducer.

**Q)The synthesis of new RNA molecules using DNA as a template is called .....**

**Answer:Term: Transcription**

- **Explanation:** Enzymatic process in which RNA polymerase synthesizes a complementary single-stranded RNA using a genomic DNA strand as a template.

**Q)..... is a process of virus-mediated transfer of genetic material from one host cell to another.**

**Answer:Term: Transduction**

- **Explanation:** Bacteriophage-mediated horizontal gene transfer mechanism in bacteria during lytic or lysogenic cycles.

**Q) cDNA is produced usually by using ..... as a template.**

**Answer:Term: mRNA (Messenger RNA)**

- **Explanation:** Processed eukaryotic mRNA possessing a 3' poly-A tail serves as the starting template for reverse transcriptase synthesis.

**Q)The polymerase chain reaction (PCR) technique is used for gene .....**

**Answer:Term: Amplification**

- **Explanation:** An *in vitro* enzymatic method that exponentially replicates target DNA sequences through automated temperature cycles.

**Q)..... is a protein synthesized by a regulator gene that binds to the operator to block transcription.**

**Answer:Term: Repressor**

- **Explanation:** Serves as a negative regulatory protein blocking RNA polymerase progression along structural genes.

**Q)Monoclonal antibodies can be produced in specialized cells by ..... technology.**

**Answer:Term: Hybridoma**

- **Explanation:** Method yielding identical monospecific antibodies targeting a single antigenic epitope continuously.

**Q) DNA finger printing is based on the presence of ..... in the human genome.**

**Answer:Term: VNTRs (Variable Number Tandem Repeats)**

- **Explanation:** Highly variable repetitive satellite DNA sequences whose unique fragment

lengths create individual genomic patterns.

**Q) ..... are small DNA sequences that can move to any position in the cell's genome.**

**Answer:Term: Transposons (or Transposable Elements)**

- **Explanation:** Mobile elements carrying transposase machinery enabling self-directed insertion into target chromosomal loci.

**Q)The amino acid coding sequences which are present in genes of eukaryotes are called .....**

**Answer:Term: Exons**

- **Explanation:** Expressed coding segments retained in mature mRNA following pre-mRNA splicing to code for functional proteins.

**Question: Define Genome.**

**Answer:** A **genome** is the complete set of genetic material (DNA or RNA in some viruses) present in an organism, containing all the necessary instructions to build, maintain, and develop that organism.

**Question: Define MHC.**

**Answer: Major Histocompatibility Complex (MHC)** is a set of cell surface proteins essential for the adaptive immune system to recognize foreign antigens, enabling T-cells to distinguish self-antigens from non-self antigens.

**Question: Define Transposon.**

**Answer:** A **transposon** (or jumping gene) is a sequence of DNA that can change its position within a genome, creating or reversing mutations and altering the cell's genetic identity and genome size.

**Question: Define Restriction Endonuclease.**

**Answer:** A **restriction endonuclease** is a bacterial enzyme that cleaves double-stranded DNA at specific recognition nucleotide sequences (restriction sites), serving as a crucial tool in recombinant DNA technology.

**Question: State whether True or False with reason: "The coding regions of a gene contains highly repetitive sequences."**

**Answer:**

- **Statement: False**
- **Reason:** The coding regions (exons) of functional genes primarily consist of unique or low-copy DNA sequences that code for specific amino acids. Highly repetitive sequences are typically present in non-coding regions (introns, heterochromatin, or intergenic spacers).

**Question: State whether True or False with reason: "In humoral immune response only B-Cells are involved."**

**Answer:**

- **Statement: False**
- **Reason:** Although B-cells produce antibodies responsible for humoral immunity, Helper T-cells (T<sub>H</sub> cells) are essential to activate B-cells for effective antibody secretion, affinity maturation, and memory cell differentiation.

**Question: State whether True or False with reason: "If a cell contains**

**only dideoxy-nucleotides, it can not replicate its DNA."**

**Answer:**

- **Statement: True**
- **Reason:** Dideoxynucleotides (ddNTPs) lack the 3'-OH group required for phosphodiester bond formation with the incoming nucleotide. Consequently, DNA polymerase cannot elongate the chain, leading to immediate termination of DNA replication.

**Question: Write briefly on Intron.**

**Answer:** An **intron** is a non-coding sequence of DNA within a gene that is transcribed into precursor mRNA (pre-mRNA) but is subsequently removed by RNA splicing before translation into protein. Exons are joined together to form the functional mature mRNA.

**Question: Write briefly on TH Cells.**

**Answer: Helper T-cells (T<sub>H</sub> cells)** are a subtype of T-lymphocytes that express CD4 surface receptors. They play a central role in adaptive immunity by recognizing antigens presented by MHC Class II molecules and releasing cytokines to activate B-cells, Cytotoxic T-cells, and macrophages.

**Question: Write briefly on Cloning Vector.**

**Answer:** A **cloning vector** is a small piece of DNA (e.g., plasmid, bacteriophage) taken from any organism into which a foreign DNA fragment can be inserted for cloning purposes. Key characteristics include an origin of replication (ori), selectable markers, and unique restriction enzyme cleavage sites.

**Question: Write briefly on TATA Box.**

**Answer:** The **TATA box** is a highly conserved DNA sequence (typically 5'-TATAAA-3') located in the promoter region of eukaryotic genes (about 25-30 base pairs upstream of the transcription start site). It serves as the binding site for RNA polymerase II and transcription factors to initiate transcription.

**Question: Differentiate between MHC I and MHC II.**

**Answer:**

| Feature                  | MHC Class I                                                             | MHC Class II                                                                   |
|--------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Expression</b>        | On all nucleated cells.                                                 | On Antigen Presenting Cells (APCs) like dendritic cells, B-cells, macrophages. |
| <b>Presents To</b>       | Cytotoxic T-cells (CD8 <sup>+</sup> T-cells).                           | Helper T-cells (CD4 <sup>+</sup> T-cells).                                     |
| <b>Source of Antigen</b> | Endogenous antigens (viral/tumor proteins synthesized inside the cell). | Exogenous antigens (pathogens phagocytosed from outside the cell).             |

**Question: Differentiate between Inducible and Repressible prokaryotic gene regulation.**

**Answer:**

| Feature                  | Inducible System (e.g., <i>lac</i> Operon)                      | Repressible System (e.g., <i>trp</i> Operon)                       |
|--------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Default State</b>     | Off (transcription normally inhibited).                         | On (transcription normally active).                                |
| <b>Effector Molecule</b> | Inducer (turns transcription ON by inactivating the repressor). | Corepressor (turns transcription OFF by binding to the repressor). |
| <b>Pathological Role</b> | Regulates catabolic pathways (e.g., degradation of lactose).    | Regulates anabolic pathways (e.g., synthesis of tryptophan).       |

**Question: Explain briefly: Nucleosome.**

**Answer:** A **nucleosome** is the fundamental structural unit of eukaryotic chromatin. It consists of approximately 146 base pairs of DNA wrapped around a protein octamer composed of two molecules each of histones H2A, H2B, H3, and H4, stabilized by histone H1.

**Question: Explain briefly: Insertional Inactivation.**

**Answer: Insertional inactivation** is a recombinant DNA technique used to screen recombinant clones. Insertion of a foreign DNA fragment into a specific restriction site within a gene (e.g., *lacZ* or antibiotic resistance gene) disrupts its function, leading to a loss of phenotype that allows easy identification of recombinants.

**Question: Explain briefly: Isochizomers.**

**Answer: Isoschizomers** are pairs of restriction endonucleases isolated from different bacterial species that recognize the exact same target DNA sequence and cleave it in the same manner (e.g., *SphI* and *NbuI* both recognize 5'-GCATGC-3').

**Question: Explain briefly: Alkaline Phosphatase.**

**Answer: Alkaline phosphatase** is an enzyme used in genetic engineering to remove the 5' phosphate group from linear DNA or RNA molecules. This prevents self-ligation (re-circularization) of vector DNA during gene cloning.

**Question: Explain briefly: Epitope and Paratope.**

**Answer:**

- **Epitope (Antigenic Determinant):** The specific molecular region of an antigen that is recognized and bound by an antibody or T-cell receptor.
- **Paratope (Antigen-Binding Site):** The specific region of an antibody's variable domain that recognizes and binds to the epitope.

### **Question: Explain briefly: Chimeric DNA.**

**Answer:** **Chimeric DNA** (or recombinant DNA) is a hybrid DNA molecule artificially constructed *in vitro* by combining DNA fragments from two or more different biological sources.

### **Question: Explain briefly: DNA Ligase.**

**Answer:** **DNA ligase** is an enzyme that repairs or joins single-stranded nicks or double-stranded breaks in DNA by catalyzing the formation of a phosphodiester bond between adjacent 3'-OH and 5'-phosphate groups.

### **Question: Explain briefly: Promoter.**

**Answer:** A **promoter** is a specific DNA sequence located upstream of a gene that provides a recognition site for RNA polymerase and regulatory transcription factors to initiate gene transcription.

### **Question: Explain briefly: Reverse Transcriptase.**

**Answer:** **Reverse transcriptase** is an RNA-dependent DNA polymerase enzyme (found in retroviruses) that synthesizes a complementary DNA (cDNA) strand using a single-stranded RNA molecule as a template.

### **Question: Explain briefly: Allergy.**

**Answer:** An **allergy** is an hypersensitive immune response triggered by normally harmless environmental substances (allergens). It is predominantly mediated by IgE antibodies, leading to mast cell degranulation and release of histamine.

### **Question: Explain briefly: Operon.**

**Answer:** An **operon** is a functional unit of genomic DNA found in prokaryotes containing a cluster of genes under the control of a single promoter and operator region, allowing coordinated gene expression and regulation.

### **Question: Explain briefly: Lac Z Gene.**

**Answer:** The **Lac Z gene** is a structural gene located within the *lac* operon of *E. coli*. It codes for the enzyme **β-galactosidase**, which hydrolyzes lactose into glucose and galactose and converts lactose to allolactose (the natural inducer of the operon).

### **Question: Explain briefly: Repetitive DNA Sequence.**

**Answer:** **Repetitive DNA sequences** are patterns of nucleic acids that occur in multiple copies throughout the genome. They are broadly classified into tandem repeats (e.g., satellite DNA, microsatellites) and interspersed repeats (e.g., LINEs, SINEs, transposons), playing roles in

chromosome structure and regulation.

**Question: Explain briefly: Cellular Component of Immune System.**

**Answer:** The **cellular components of the immune system** include white blood cells (leukocytes) responsible for immune defense. These encompass innate immune cells (macrophages, neutrophils, dendritic cells, NK cells) and adaptive immune cells (B-lymphocytes and T-lymphocytes).

**Question: Explain briefly: Somatic Cell Hybridization.**

**Answer:** **Somatic cell hybridization** is an *in vitro* technique where somatic cells from two different species (e.g., human and mouse) are fused using fusogens like polyethylene glycol (PEG) or Sendai virus to form a hybrid cell, widely used for chromosome mapping and monoclonal antibody production.

**Question: Explain briefly: Histone Protein.**

**Answer:** **Histone proteins** are highly conserved, basic (positively charged) proteins rich in lysine and arginine residues that package eukaryotic DNA into structural units called nucleosomes, playing a pivotal role in chromatin compaction and gene regulation.

**Question: Explain briefly: cDNA.**

**Answer:** **cDNA (Complementary DNA)** is synthetic DNA synthesized from a mature mRNA template via reverse transcription. Because it lacks non-coding introns, cDNA is extensively used in gene cloning, expression systems, and genetic engineering.

**Question: Explain briefly: DNA Polymerase.**

**Answer:** **DNA polymerase** is an enzyme essential for DNA replication that catalyzes the synthesis of new DNA strands by adding complementary deoxyribonucleotides to the 3'-OH end of a growing primer strand in the 5' to 3' direction.

# 1. Restriction Enzymes

**Definition :-** Restriction enzymes, formally known as restriction endonucleases, are specialized enzymes that cleave double-stranded DNA at specific nucleotide sequences called recognition sites. Discovered by Werner Arber, Daniel Nathans, and Hamilton Smith, these enzymes naturally occur in bacteria, where they act as a primitive immune defense mechanism (the restriction-modification system) against invading bacteriophages by digesting viral DNA.

**Classification and Types** Restriction endonucleases are categorized into four major types (Types I, II, III, and IV) based on their subunit structure, cofactor requirements, and cleavage mechanism:

- **Type I:** Require S-adenosylmethionine, ATP, and  $Mg^{2+}$ . They cleave DNA at random sites far from the recognition sequence (often >1000 bp away).
- **Type II:** The most crucial type for molecular biology and recombinant DNA technology. They require only  $Mg^{2+}$  as a cofactor and cut DNA precisely within or immediately adjacent to their specific palindromic recognition site.
- **Type III:** Require ATP and  $Mg^{2+}$  and cut DNA about 25 to 27 base pairs away from the recognition site.
- **Type IV:** Cleave only methylated DNA.

**Mechanism of Action and Cleavage Patterns** Type II enzymes recognize palindromic sequences (sequences that read the same on both complementary strands in the 5'→3' direction, such as 5'-GAATTC-3' for *EcoRI*). When cutting, they generate two types of ends:

- **Sticky (Cohesive) Ends:** Staggered cuts made in the sugar-phosphate backbone leave short, single-stranded overhangs. Examples include *EcoRI* and *HindIII*. These ends readily anneal with complementary sequences via hydrogen bonding.
- **Blunt Ends:** Flush cuts directly down the axis of symmetry, leaving no single-stranded overhangs. An example is *SmaI*.

**Applications in Biotechnology** Restriction endonucleases are essential tools for construct design in gene cloning, gene isolation, restriction mapping, recombinant plasmid construction, and Restriction Fragment Length Polymorphism (RFLP) analysis.

## 2. Scope of Genetic Engineering

**Definition :-** Genetic engineering, or recombinant DNA technology, involves the deliberate, direct manipulation of an organism's genome using biotechnology. It encompasses the isolation, manipulation, recombination, and re-introduction of modified genes into host organisms to alter their phenotypic characteristics.

**Applications in Medicine and Healthcare**

- **Therapeutics Production:** Synthesis of human therapeutic proteins like recombinant insulin (humulin), human growth hormone, clotting factor VIII, and erythropoietin using microbial host systems.
- **Gene Therapy:** Correction of defective genes responsible for hereditary diseases such as Severe Combined Immunodeficiency (SCID) and cystic fibrosis.
- **Vaccine Development:** Production of safe subunit and recombinant vaccines, such as the Hepatitis B vaccine, eliminating the risks associated with live attenuated pathogens.

**Applications in Agriculture and Animal Husbandry**

- **Transgenic Crops:** Development of Genetically Modified (GM) crops engineered for biotic stress resistance (e.g., Bt cotton resistant to bollworms) and abiotic stress tolerance (drought, salinity).
- **Nutritional Enhancement:** Biofortification of food crops, such as Golden Rice engineered to synthesize  $\beta$ -carotene (Vitamin A precursor).
- **Transgenic Animals:** Engineering livestock for higher milk production, leaner meat, disease resistance, and bioreactor applications (pharmaceutical production in milk).

#### Environmental and Industrial Applications

- **Bioremediation:** Engineering microbes (e.g., *Pseudomonas putida*) to degrade complex pollutants, oil spills, toxic heavy metals, and plastic waste.
- **Industrial Enzymes:** Large-scale production of enzymes like amylases, proteases, and cellulases for detergents, food processing, and biofuel production.

### 3. Hazards of Recombinant DNA Technology

While recombinant DNA (rDNA) technology offers immense scientific and medical breakthroughs, it presents potential biological, environmental, and ethical hazards. Regulatory authorities worldwide enforce biosafety levels (BSL-1 to BSL-4) to contain risks associated with genetically modified organisms (GMOs).

#### Ecological and Environmental Hazards

- **Gene Flow and Superweeds:** The accidental transfer of herbicide-resistance or pest-resistance transgenes from GM crops to wild relatives through cross-pollination can create invasive "superweeds."
- **Impact on Non-Target Organisms:** Expression of insecticidal proteins (like Cry toxins) in transgenic plants may adversely affect beneficial non-target insects, such as pollinators (bees) and soil microflora.
- **Loss of Biodiversity:** Widespread monoculture of uniform transgenic crop varieties reduces genetic diversity, making agricultural ecosystems vulnerable to novel pests or environmental changes.

#### Health Hazards to Humans and Animals

- **Allergenicity and Toxicity:** Introduction of novel foreign proteins into food crops can trigger unforeseen allergic reactions or metabolic toxicity in consumers.
- **Antibiotic Resistance Marker Transfer:** The use of antibiotic resistance genes as selection markers in rDNA vectors carries a theoretical risk of horizontal gene transfer to human intestinal bacteria, contributing to global antibiotic resistance.

#### Biological Safety and Dual-Use Concerns

- **Creation of Superpathogens:** Unintended or deliberate modification of infectious pathogens could produce drug-resistant, hyper-virulent strains capable of bypassing human immunity.
- **Bioterrorism Risks:** The accessibility of synthetic biology tools raises biosafety concerns regarding the production of biological weapons or weaponized pathogens.

### 4. Lac Operon

**Definition and Historic Background** The *lac* operon (lactose operon) is an inducible genetic system in *Escherichia coli* responsible for the transport and catabolism of lactose. The mechanism was elucidated by François Jacob and Jacques Monod in 1961, providing the

foundational model for gene regulation in prokaryotes.

**Structural and Regulatory Components** The operon consists of structural genes, a promoter, an operator, and a distinct regulatory gene:

- **Structural Genes:**
  - ***lacZ***: Encodes  $\beta$ -galactosidase, an enzyme that hydrolyzes lactose into glucose and galactose.
  - ***lacY***: Encodes  $\beta$ -galactoside permease, a membrane-bound protein that facilitates lactose entry into the cell.
  - ***lacA***: Encodes  $\beta$ -galactoside transacetylase, which transfers an acetyl group to  $\beta$ -galactosides.
- **Promoter (*lacP*)**: Binding site for RNA polymerase to initiate transcription.
- **Operator (*lacO*)**: Sequence overlapping the promoter where the repressor protein binds to block transcription.
- **Regulatory Gene (*lacI*)**: Continuously encodes the *lac* repressor protein.

#### Mechanism of Regulation

- **In the Absence of Lactose (Off State)**: The *lacI* gene produces an active repressor tetramer that binds tightly to the operator region (*lacO*). This physical blockage prevents RNA polymerase from transcribing the structural genes, conserving cellular energy.
- **In the Presence of Lactose (On State)**: Lactose is converted into allolactose, which acts as the inducer. Allolactose binds to the *lac* repressor, causing a conformational change that lowers its affinity for the operator. The repressor dissociates from the operator, allowing RNA polymerase to transcribe *lacZ*, *lacY*, and *lacA* into a single polycistronic mRNA.

**Catabolite Repression** Even in the presence of lactose, if glucose is available, the operon remains largely inactive. High glucose levels lower cyclic AMP (cAMP) levels. Without the cAMP-CAP (Catabolite Activator Protein) complex bound upstream of the promoter, RNA polymerase cannot efficiently initiate transcription, ensuring glucose is utilized first.

## 5. Cell Hybrids

**Definition** :-Cell hybrids, or somatic cell hybrids, are engineered cells formed by fusing two genetically distinct somatic cells (often from different tissue types or different species, such as human and mouse) into a single combined cell. The resulting cell contains nuclear genomes from both parent cells.

#### Process of Somatic Cell Hybridization

- **Induction of Fusion**: Somatic cells grown in tissue culture are induced to fuse using fusogenic agents such as Polyethylene Glycol (PEG) or inactivated Sendai virus. Electroporation can also be applied.
- **Formation of Heterokaryon**: Initial fusion yields a multinucleated cell containing two separate nuclei within a common cytoplasm, termed a heterokaryon.
- **Formation of Synekaryon**: During subsequent mitotic divisions, the nuclear membranes dissolve, and the two sets of chromosomes merge into a single unified nucleus, producing a true hybrid cell or synekaryon.

**Selective Systems (HAT Selection)** To isolate hybrid cells from unfused parent cells, a selection medium known as HAT medium (Hypoxanthine, Aminopterin, Thymidine) is used:

- Aminopterin blocks the primary *de novo* pathway of nucleotide synthesis.
- Hybrid cells survive by utilizing the salvage pathway, provided they inherit enzyme

functionality from both parents—such as Hypoxanthine-Guanine Phosphoribosyltransferase (HGPRT<sup>+</sup>) from one parent and Thymidine Kinase (TK<sup>+</sup>) from the other.

**Applications** Somatic cell hybrids are instrumental in gene mapping, determining human chromosomal gene locations (as human-mouse hybrids selectively lose human chromosomes over time), studying gene expression control, and producing monoclonal antibodies via hybridoma technology.

## 6. End Labeling of DNA

**Definition and Concept** End labeling of DNA is a molecular technique used to attach a detectable label—typically a radioactive isotope (such as <sup>32</sup>P) or a non-radioactive tag (such as biotin or digoxigenin)—specifically to either the 5' or 3' terminus of a DNA strand. This selective labeling is vital for tracking, detecting, and analyzing DNA fragments in molecular biology assays.

**5' End Labeling Technique** 5' end labeling is performed using the enzyme T4 Polynucleotide Kinase (T4 PNK).

- The existing non-radioactive phosphate group at the 5' end of the DNA molecule is first removed using an alkaline phosphatase enzyme (e.g., Calf Intestinal Phosphatase, CIP).
- T4 PNK is then introduced along with [gamma-<sup>32</sup>P] ATP. The enzyme transfers the labeled gamma-phosphate group from ATP onto the free 5' hydroxyl group of the target DNA strand.

**3' End Labeling Technique** 3' end labeling can be accomplished using Terminal Deoxynucleotidyl Transferase (TdT) or the Klenow fragment of DNA Polymerase I.

- TdT adds labeled nucleotides (such as [alpha-<sup>32</sup>P] dNTPs or dideoxynucleotides) sequentially to the 3' hydroxyl end of single- or double-stranded DNA in a template-independent manner.
- Alternatively, for DNA fragments with staggered 5' overhangs, the Klenow fragment fills in the overhang using labeled dNTPs to yield 3' labeled ends.

**Applications** End-labeled DNA probes are widely used in restriction site mapping, Maxam-Gilbert DNA sequencing, DNA footprinting assays, gel mobility shift assays (EMSA), and Southern blot hybridization.

## 7. Microinjection of Genes into Animal Oocytes

**Definition and Technique** Microinjection is a direct, physical method of gene transfer in which foreign recombinant DNA is directly injected into the nucleus or cytoplasm of animal oocytes, zygotes, or early embryos using a specialized glass microcapillary pipette under high-power microscopic observation.

**Procedure and Methodology**

- **Superovulation and Egg Retrieval:** Female donor animals are treated with gonadotropin hormones to induce superovulation. Unfertilized oocytes or single-cell zygotes are collected from the oviducts.
- **Microinjection Setup:** The retrieved oocyte is immobilized on a holding pipette using gentle suction under an inverted micromanipulator microscope.
- **Injection of DNA:** A microscopic glass needle with a tip diameter of 0.5 to 1.5 micrometers, pre-loaded with a solution of the gene of interest, pierces the oocyte plasma

membrane. For fertilized eggs, the DNA is injected directly into the male pronucleus before nuclear fusion occurs, as it is larger and more visible.

- **Implantation:** The microinjected oocytes or zygotes are cultured *in vitro* to viable cleavage stages and then surgically transferred into the oviducts or uterus of pseudopregnant surrogate mothers.

#### **Advantages and Limitations**

- **Advantages:** High transformation efficiency per injected cell; precise delivery of measured DNA volumes; does not require specialized chemical vectors or viral carriers.
- **Limitations:** Requires expensive equipment and technical skill; low overall success rate regarding stable transgene integration into the host genome; potential for mechanical damage to the egg cell.

## **8. Major Histocompatibility Complex (MHC)**

**Definition and Biological Significance** The Major Histocompatibility Complex (MHC) is a dense, highly polymorphic cluster of cell-surface antigen-presenting proteins encoded by a large gene complex. In humans, the MHC is known as the Human Leukocyte Antigen (HLA) complex, located on short arm of chromosome 6, while in mice, it is called the H-2 complex (chromosome 17). MHC molecules are central to self/non-self recognition and adaptive immune responses.

#### **Classes of MHC Molecules**

- **MHC Class I:**
  - **Structure:** Consists of a heavy  $\alpha$  transmembrane chain non-covalently associated with a smaller, invariant  $\beta_2$ -microglobulin protein.
  - **Tissue Distribution:** Expressed on the surface of virtually all nucleated somatic cells and platelets.
  - **Function:** Presents endogenous antigens (derived from intracellular pathogens like viruses or tumor proteins) to Cytotoxic T Lymphocytes (CD8<sup>+</sup> T cells).
- **MHC Class II:**
  - **Structure:** Heterodimer comprising two non-covalently linked transmembrane glycoprotein chains ( $\alpha$  and  $\beta$ ).
  - **Tissue Distribution:** Restricted primarily to professional Antigen-Presenting Cells (APCs), including dendritic cells, macrophages, and B lymphocytes.
  - **Function:** Presents exogenous antigens (engulfed extracellular pathogens) to Helper T Lymphocytes (CD4<sup>+</sup> T cells).
- **MHC Class III:** Encodes non-membrane proteins involved in immune cascades, such as complement factors (C2, C4, Factor B) and inflammatory cytokines (TNF- $\alpha$ , lymphotoxin).

**Polymorphism and Clinical Importance** MHC genes are the most polymorphic genes known in vertebrates. This immense genetic diversity ensures that a population can present a vast array of microbial peptides. MHC compatibility is a major determinant in tissue and organ transplantation, where mismatched MHC alleles trigger graft rejection.

## **9. DNA Microinjection**

**Definition and Principles** DNA microinjection is a mechanical gene-delivery technique used to introduce foreign genetic material directly into host living cells—such as somatic cells, cultured lines, plant protoplasts, and embryonic cells—using fine microcapillary needles operated via

high-precision micromanipulators.

#### Instrumentation Required

- **Micromanipulators:** Precision mechanical devices that convert broad hand movements into microscopic movements of the injection tip.
- **Inverted Microscope:** Outfitted with phase-contrast or differential interference contrast optics to provide clear visualization of cell structures and nuclei.
- **Micro-injector System:** Pneumatic or hydraulic pressure systems capable of dispensing nanoliter-to-picoliter volumes of liquid DNA solutions.
- **Micro-pipettes:** Micro-holding pipettes to anchor the target cell via gentle suction, and micro-injection needles with sub-micron tip diameters.

#### Procedure

1. The recipient target cells are anchored on a glass coverslip or held in position by a holding pipette.
2. The fine micro-injection needle containing the purified linear recombinant DNA construct is advanced carefully through the cell membrane and nuclear envelope.
3. A controlled pulse of pressure delivers the micro-volume of DNA solution directly into the nucleus, following which the needle is withdrawn.

**Applications and Impact** DNA microinjection is widely used for producing transgenic animals, studying nuclear gene regulation, investigating developmental biology pathways, and performing gene function analysis *in vivo*.

## 10. Reverse Transcriptase

**Definition and Discovery** Reverse transcriptase, also known as RNA-directed DNA polymerase, is an enzyme that catalyzes the synthesis of complementary DNA (cDNA) using a single-stranded RNA molecule as a template. Discovered independently by Howard Temin and David Baltimore in 1970, this enzyme challenged the classical Central Dogma of molecular biology (DNA  $\rightarrow$  RNA  $\rightarrow$  Protein).

**Enzymatic Activities** Reverse transcriptase possesses three distinct enzymatic functions:

- **RNA-Dependent DNA Polymerase:** Synthesizes a single-stranded DNA strand complementary to an RNA template in the 5'  $\rightarrow$  3' direction, requiring a short primer with a free 3'-OH group.
- **RNase H Activity:** Degrades the original RNA strand in the resulting RNA-DNA hybrid, leaving the single-stranded cDNA intact.
- **DNA-Dependent DNA Polymerase:** Synthesizes the second strand of DNA using the newly generated single-stranded cDNA as a template, producing double-stranded cDNA.

**Biological Occurrence** Reverse transcriptase naturally occurs in retroviruses (such as HIV and Rous Sarcoma Virus), retrotransposons (like LINEs and LINE-like elements), and human telomerase enzymes (which maintain chromosome end lengths).

**Applications in Molecular Biology** Reverse transcriptase is a core enzyme in biotechnology. It is used in Reverse Transcription Polymerase Chain Reaction (RT-PCR) to quantify mRNA levels, in constructing cDNA libraries representing actively transcribed genes, and in generating molecular probes from cellular transcripts.

## 11. Transduction in Bacteria

**Definition and Historical Discovery** Transduction is a process of horizontal gene transfer in

bacteria where bacterial genomic DNA is transferred from a donor bacterium to a recipient bacterium through a bacteriophage (bacterial virus) vector. The mechanism was discovered by Joshua Lederberg and Norton Zinder in 1952 using *Salmonella typhimurium* and bacteriophage P22.

**Types of Transduction** Transduction occurs via two primary mechanisms based on the lifecycle of the participating bacteriophage:

### 1. Generalized Transduction

- **Mechanism:** Occurs during the lytic cycle of virulent or temperate phages (e.g., phage P22 or P1).
- **Process:** As the host bacterial chromosome is degraded during viral replication, random fragments of bacterial host DNA are mistakenly packaged into developing phage heads instead of viral DNA.
- **Outcome:** The resulting "transducing phage" infects a new recipient bacterium, injecting the donor bacterial DNA segment, which can integrate into the recipient chromosome via homologous recombination. Any gene on the host chromosome has an equal probability of being transferred.

### 2. Specialized (Restricted) Transduction

- **Mechanism:** Mediated by lysogenic (temperate) phages (e.g., Lambda phage in *E. coli*) during the transition from the lysogenic to the lytic cycle.
- **Process:** The prophage integrates into a specific site on the bacterial chromosome (e.g., the *att* site near the *gal* and *bio* operons in *E. coli*). Upon induction, imprecise excision of the prophage occurs, picking up adjacent bacterial genes along with a portion of the phage genome.
- **Outcome:** Only specific bacterial genes located adjacent to the prophage integration site are transferred to recipient cells.

**Significance** Transduction plays a vital role in bacterial evolution, transferring antibiotic resistance genes and virulence factors across species, and serves as a classic tool for bacterial genetic mapping.

## 12. RNA Splicing

**Definition and Biological Significance** RNA splicing is a critical post-transcriptional modification step in eukaryotic gene expression. During this process, non-coding sequences called introns are precisely removed from the precursor messenger RNA (pre-mRNA), and coding sequences called exons are joined together to yield a mature mRNA molecule capable of being translated into protein.

**The Splicing Machinery (Spliceosome)** RNA splicing is carried out within the nucleus by a large ribonucleoprotein complex called the spliceosome.

- The spliceosome is assembled from five small nuclear ribonucleoproteins (snRNPs, pronounced "snurps"): U1, U2, U4, U5, and U6, alongside additional non-snRNP protein factors.
- Each snRNP contains small nuclear RNA (snRNA) molecules that recognize specific consensus sequences at the intron boundaries.

**Mechanism of Splicing** The removal of an intron occurs through two sequential transesterification reactions:

1. **5' Splice Site Recognition:** The U1 snRNP binds to the conserved 5' splice site (GU sequence), and the U2 snRNP binds to the internal branch point sequence (containing an

invariant adenine).

2. **First Transesterification:** The 2'-OH group of the branch-point adenine performs a nucleophilic attack on the phosphate group at the 5' splice site, breaking the phosphodiester bond and forming a branched "lariat" intron structure.
3. **Second Transesterification:** The free 3'-OH group of the cleaved upstream exon attacks the phosphate at the 3' splice site (AG sequence). This joins the two exons together and releases the excised intron as a lariat, which is subsequently degraded.

**Alternative Splicing** Alternative splicing allows different combinations of exons from a single pre-mRNA to be included or excluded. This single gene mechanism enables eukaryotic cells to produce multiple distinct protein isoforms from a single gene, greatly expanding proteomic diversity.

## 13. Transfection

**Definition and Overview** Transfection refers to the process of deliberately introducing nucleic acids (DNA, RNA, or siRNA) into eukaryotic cells—specifically animal cells—using non-viral physical, chemical, or biological procedures. (The term "transformation" is generally reserved for bacteria and fungi, or malignant conversion in animal cells).

**Major Transfection Methods** Transfection protocols are broadly classified into chemical and physical techniques:

- **Chemical Methods:**
  - **Calcium Phosphate Co-precipitation:** DNA is mixed with calcium chloride and phosphate buffer to form insoluble calcium phosphate-DNA micro-precipitates. Cells take up these particles via endocytosis.
  - **Lipofection (Liposome-Mediated Transfection):** Positively charged (cationic) lipids condense negatively charged nucleic acids into liposomes. These lipid complexes fuse with the negatively charged plasma membrane, releasing DNA into the cytoplasm.
  - **DEAE-Dextran Transfection:** Cationic polymers bind negatively charged DNA molecules, promoting interaction with the cell surface and endocytic internal uptake.
- **Physical Methods:**
  - **Electroporation:** Short, high-voltage electrical pulses create transient micro-pores in the cell membrane, allowing nucleic acids to diffuse into the cytoplasm.
  - **Microinjection:** Direct mechanical injection of DNA into the cell nucleus using capillary needles.
  - **Biolistics (Gene Gun):** Heavy metal microparticles (gold/tungsten) coated with DNA are accelerated at high velocity into target tissue.

### Types of Expression

- **Transient Transfection:** The introduced DNA remains as an extrachromosomal element inside the nucleus and is lost during cell division over time.
- **Stable Transfection:** Transfected DNA integrates stably into the host cell's chromosomal genome, enabling long-term expression and inheritance by daughter cells.

## 14. Transgenic Animals and Plants

**Definition** Transgenic organisms are animals or plants whose genomes have been stably modified by the artificial introduction of exogenous foreign DNA (transgenes) from another

species using recombinant DNA techniques. Transgenes are incorporated into the germline so they can be passed on to offspring.

### Transgenic Animals

- **Methods of Creation:** Generated primarily through microinjection of DNA into fertilized eggs (pronuclear injection), retroviral vector infection of early embryos, or somatic cell nuclear transfer (cloning) using genetically altered somatic nuclei.
- **Key Examples and Applications:**
  - **PharmAnimals:** Transgenic livestock (e.g., sheep, goats) engineered to secrete human therapeutic proteins (such as alpha<sub>1</sub>-antitrypsin or tissue plasminogen activator) in their milk, a process termed "gene farming" or molecular pharming.
  - **Disease Models:** Knockout mice (e.g., OncoMouse) designed to carry human genes to study cancer, Alzheimer's, and metabolic diseases.

### Transgenic Plants

- **Methods of Creation:** Generated mainly using *Agrobacterium tumefaciens*-mediated transformation (utilizing the modified Ti plasmid) or physical gene delivery techniques like biolistic particle bombardment.
- **Key Examples and Applications:**
  - **Insect-Resistant Plants:** Bt crops (Bt cotton, Bt corn) engineered with the *Cry* gene from *Bacillus thuringiensis* to produce insecticidal endotoxins.
  - **Herbicide-Tolerant Crops:** "Roundup Ready" soybeans and corn carrying resistance to glyphosate herbicides, simplifying weed control.
  - **Nutrient Enriched Plants:** Golden Rice modified with carotenoid biosynthesis genes to express beta-carotene in grain endosperm.

## 15. Monoclonal Antibodies

**Definition and Characterization** Monoclonal antibodies (mAbs) are monospecific, highly uniform antibodies produced by a single clone of B lymphocytes. Because they originate from a single B-cell clone, all antibody molecules in a monoclonal preparation are identical in structure, binding affinity, and recognize the exact same antigenic epitope.

**Production via Hybridoma Technology** Developed by Georges Köhler and César Milstein in 1975, hybridoma technology enables the continuous, large-scale production of monoclonal antibodies:

1. **Immunization:** A mouse is injected with a specific target antigen to stimulate an immune response.
2. **Isolation of B-Cells:** Plasma B lymphocytes producing the desired antibody are harvested from the mouse spleen.
3. **Cell Fusion:** The harvested B cells (which have a finite lifespan) are fused with immortalized myeloma cells (cancerous B cells lacking the HGPRT enzyme) using Polyethylene Glycol (PEG).
4. **Selection:** The resulting cell mixture is cultured in HAT selection medium. Unfused spleen cells die off naturally, while unfused myeloma cells die because they lack HGPRT. Only fused **hybridoma cells** survive, combining immortality with continuous antibody production.
5. **Screening and Cloning:** Hybridomas are screened (via ELISA) for specific antibody production, and single positive clones are isolated and expanded *in vitro* or *in vivo*.

### Applications

- **Diagnostic Tools:** Used in pregnancy test kits (detecting hCG), enzyme-linked immunosorbent assays (ELISA), western blotting, and blood typing.
- **Therapeutic Agents:** Directed cancer therapies (e.g., Trastuzumab/Herceptin for breast cancer, Rituximab for lymphoma) and treatments for autoimmune diseases (e.g., Infliximab for rheumatoid arthritis).

## 16. Southern Blot Hybridization

**Definition and Origin** Southern blot hybridization is a molecular biology laboratory technique used to detect the presence, size, and quantity of a specific DNA sequence within a complex mixture of DNA samples. Developed by Sir Edwin Southern in 1975, the technique combines gel electrophoresis with nucleic acid hybridization using labeled probes.

### Step-by-Step Procedure

1. **DNA Digestion:** High-molecular-weight genomic DNA is cleaved into smaller fragments using specific Type II restriction endonucleases.
2. **Gel Electrophoresis:** The resulting DNA fragments are separated according to molecular size on an agarose gel using electric current.
3. **Denaturation:** The gel is treated with an alkaline solution (such as sodium hydroxide) to denature double-stranded DNA into single-stranded DNA fragments.
4. **Blotting (Transfer):** Single-stranded DNA fragments are transferred from the gel onto a solid capillary membrane (nitrocellulose or nylon) via capillary action or electroblotting, maintaining the exact spatial arrangement from the gel.
5. **Immobilization:** The transferred DNA is permanently fixed to the membrane using heat baking (80 °C) or UV cross-linking.
6. **Hybridization:** The membrane is incubated with a labeled single-stranded DNA or RNA probe (radioactive or fluorescent) complementary to the target sequence.
7. **Detection:** Unbound probe is washed away, and the hybridized probe-target complexes are visualized using autoradiography or chemiluminescent imaging.

**Applications** Southern blotting is used in restriction fragment length polymorphism (RFLP) analysis, viral gene integration detection, diagnosis of genetic disorders, and DNA fingerprinting.

## 17. Interferons

**Definition and Properties** Interferons (IFNs) are a natural group of signaling proteins, specifically cytokines, synthesized and released by host cells (such as white blood cells, fibroblasts, and T cells) in response to viral infections, parasites, and tumor development. They function as potent antiviral and immunomodulatory signaling agents that establish an "antiviral state" in surrounding uninfected cells.

**Classification of Interferons** Interferons are grouped into three main classes based on receptor specificity and structural properties:

- **Type I Interferons:** Includes **Interferon-alpha (IFN-alpha)** (produced primarily by leukocytes) and **Interferon-beta (IFN-beta)** (produced by fibroblasts). They bind to the heterodimeric IFNAR receptor complex and exert potent antiviral and antiproliferative activities.
- **Type II Interferon:** Comprises **Interferon-gamma (IFN-gamma)** (produced by activated T cells and Natural Killer cells). It acts primarily as an immunomodulator, activating macrophages, increasing MHC Class I and II antigen expression, and enhancing

cell-mediated immunity.

- **Type III Interferons:** Comprises **Interferon-lambda (IFN-lambda)** isoforms, which act through distinct receptors found predominantly on epithelial tissues.

#### **Mechanism of Antiviral Action**

1. An infected cell detects viral nucleic acids and secretes interferons into the extracellular environment.
2. Interferons bind to specific surface receptors on neighboring, uninfected target cells, activating the intracellular JAK-STAT signaling pathway.
3. This pathway induces transcription of hundreds of Interferon-Stimulated Genes (ISGs).
4. ISG expression leads to key antiviral enzymes:
  - **2',5'-Oligoadenylate Synthetase:** Activates RNase L to degrade viral and host RNA.
  - **Protein Kinase R (PKR):** Phosphorylates initiation factor eIF-2alpha, halting protein translation and preventing viral replication.

**Clinical Applications** Recombinant interferons are used medically to treat chronic viral infections (such as Hepatitis B and C), multiple sclerosis (IFN-beta), and malignancies like hairy-cell leukemia and Kaposi's sarcoma.

## **18. Transposons**

**Definition and Discovery** Transposons, also known as transposable elements (TEs) or "jumping genes," are distinct DNA sequences capable of changing their location within a genome. Discovered by Barbara McClintock in the 1940s during her genetic studies on Indian corn (*Zea mays*)—work that earned her the Nobel Prize in 1983—transposons are key drivers of genomic instability and evolution.

**Classes of Transposable Elements** Transposons are categorized into two major classes according to their mechanism of transposition:

- **Class I: Retrotransposons (Copy-and-Paste)**
  - **Mechanism:** They transcribe their DNA into an RNA intermediate, which is then reverse-transcribed back into a complementary DNA strand by reverse transcriptase before integrating into a new genomic locus.
  - **Types:** Include Long Terminal Repeat (LTR) retrotransposons and Non-LTR retrotransposons (such as LINEs and SINEs, e.g., *Alu* elements in humans).
- **Class II: DNA Transposons (Cut-and-Paste)**
  - **Mechanism:** They transpose directly without an RNA intermediate. The encoded enzyme **transposase** recognizes inverted terminal repeats (ITRs) flanking the transposon, excises the element from its original site, and inserts it into a target genomic location.
  - **Examples:** The *Ac/Ds* system in maize, *P* elements in *Drosophila*, and bacterial insertion sequences (IS elements).

#### **Genetic Consequences and Significance**

- **Mutagenesis:** Integration of a transposon into a structural gene can disrupt reading frames, causing gene inactivation or altered regulatory expression.
- **Genome Expansion:** Transposable elements constitute a large portion of eukaryotic genomes (nearly 45-50% of the human genome).
- **Evolutionary Role:** They facilitate gene duplication, chromosome rearrangements, and horizontal gene transfer (such as spreading antibiotic resistance cassette genes in

bacterial transposons).

## 19. Microinjection of Genes into Animal Cells

**Definition and Scope** Microinjection into somatic or cultured animal cells involves the direct mechanical delivery of cloned genetic constructs (DNA or RNA) or regulatory proteins into individual cultured cells. Unlike whole-oocyte microinjection, this approach is typically used to study gene expression, cell cycle regulation, signaling pathways, and short-term cellular responses *in vitro*.

### Methodological Workflow

- **Cell Preparation:** Target host animal cells (such as NIH 3T3 fibroblasts or CHO cells) are grown on glass-bottom culture dishes coated with cell-adhesion substrates (like fibronectin or poly-L-lysine).
- **Equipment Calibration:** Cells are visualized under an inverted microscope equipped with high-contrast optics. The micro-injection needle is adjusted using a micromanipulator to target the nucleus (for DNA transformation) or the cytoplasm (for mRNA or protein delivery).
- **Delivery:** Micro-pressures are applied to inject precise fluid volumes ( $10^{-14}$  to  $10^{-15}$  liters) per target cell.
- **Post-Injection Analysis:** Cells are incubated and subsequently evaluated using fluorescence microscopy, immunostaining, or reporter gene assays (e.g., GFP detection).

### Advantages and Limitations

- **Advantages:** High transformation rates per injected cell; avoids toxicity associated with chemical transfection reagents; allows precise targeting of either cytoplasmic or nuclear compartments.
- **Limitations:** Low throughput (cells must be injected one by one); requires high technical expertise; can cause mechanical stress or death in fragile host cells.

## 20. Gene Regulation in Heterokaryons

**Definition of Heterokaryon** A heterokaryon is a multinucleated cell produced by the experimental fusion of two genetically distinct somatic cells (e.g., a human cell and a mouse cell, or a differentiated cell and an undifferentiated cell) whose plasma membranes merge while their individual nuclei remain intact and spatially separate within the shared cytoplasm.

**Principles of Gene Regulation Studies** Heterokaryons provide a valuable biological system for studying how diffusible cytoplasmic factors (trans-acting factors) regulate gene expression, nuclear reactivation, and differentiation without requiring genomic fusion or chromosome loss.

### Key Experimental Insights

- **Reactivation of Dormant Nuclei:** When an inactive nucleus (such as an avian erythrocyte nucleus, which is transcriptionally silent) is fused with a actively dividing cell (like a HeLa cell), the silent nucleus expands, re-initiates RNA synthesis, and resumes DNA replication. This demonstrates that cytoplasm from the active cell contains trans-acting signals that reactivate dormant chromatin.
- **Dominance of Regulatory Factors:** Fusing a differentiated cell (e.g., a muscle cell producing specific proteins) with a non-muscle cell showed that cytoplasmic trans-acting factors can selectively induce or silence cell-type-specific genes across species boundaries.

**Significance** Heterokaryon experiments demonstrated that gene expression is dynamic and continuously regulated by trans-acting proteins (transcription factors, chromatin remodelers) circulating in the cytoplasm, laying the foundation for modern nuclear reprogramming and stem cell research.

## 21. Repetitive DNA Sequences

**Definition and Overview** Repetitive DNA sequences refer to regions of nucleic acid sequences present in multiple copies throughout a genome. While prokaryotes possess mostly unique, non-repetitive coding DNA, eukaryotic genomes contain substantial proportions of repetitive non-coding DNA—making up over 50% of the human genome.

**Classification of Repetitive DNA** Repetitive DNA is broadly categorized into two major classes based on structural organization:

- **1. Tandemly Repeated DNA:**
  - Repeated units are arranged head-to-tail in adjacent blocks.
  - **Satellite DNA:** Large arrays of repetitive units (100 bp to thousands of base pairs) spanning megabases, located mainly near heterochromatic regions like centromeres and telomeres.
  - **Minisatellites (VNTRs):** Repeat units of 10 to 100 base pairs, spanning 0.5 to 20 kb. Highly polymorphic between individuals, serving as the basis for traditional DNA fingerprinting.
  - **Microsatellites (STRs):** Short repeat sequences of 2 to 6 base pairs (e.g., CACACACA), spanning less than 100 bp. Used in modern forensic profiling.
- **2. Interspersed Repetitive DNA:**
  - Repetitive sequences distributed individually throughout the genome, primarily derived from transposable elements.
  - **LINEs (Long Interspersed Nuclear Elements):** Autonomous retrotransposons (~6-8 kb long), such as LINE-1 in humans.
  - **SINEs (Short Interspersed Nuclear Elements):** Non-autonomous retrotransposons (~100-400 bp long), such as the human *Alu* element.

**Biological Significance** Repetitive DNA plays important roles in chromosome structure, centromere assembly, telomeric protection, nuclear organization, meiotic crossing over, and genome evolution.

## 22. DNA Fingerprinting

**Definition and Origin** DNA fingerprinting, also known as DNA profiling or typing, is a molecular technique used to identify individuals based on unique patterns in their genomic DNA.

Developed by Sir Alec Jeffreys in 1984, the method capitalizes on hypervariable regions of non-coding repetitive DNA that differ significantly between individuals (with the exception of monozygotic twins).

**Genetic Basis** The technique relies on Variable Number Tandem Repeats (VNTRs) and Short Tandem Repeats (STRs). Because the number of repeat units at a given genomic locus varies widely across a population due to high mutation rates, an individual's combination of loci yields a unique genetic profile.

### Step-by-Step Procedure

1. **Sample Collection & Extraction:** Genomic DNA is isolated from biological samples such

as blood, saliva, hair follicles, semen, or tissue.

2. **Amplification / Digestion:** Target repetitive loci are amplified using Polymerase Chain Reaction (PCR) with specific primers, or digested with restriction endonucleases.
3. **Separation:** Amplified fragments are separated by size using gel or capillary electrophoresis.
4. **Transfer and Hybridization (Traditional RFLP Method):** DNA is blotted onto a nylon membrane and hybridized with labeled VNTR probes.
5. **Visualization:** Banding patterns are visualized using autoradiography or fluorescent detection systems, producing a pattern resembling a barcode.

#### Applications

- **Forensic Investigations:** Matching suspect DNA to biological evidence left at crime scenes.
- **Paternity and Kinship Testing:** Establishing biological relationships in disputed parentage cases.
- **Conservation Biology:** Tracking genetic diversity and preventing illegal trafficking of endangered species.

## 23. Allergy

**Definition and Fundamental Concept** An allergy, or hypersensitivity Type I reaction, is an exaggerated, inappropriate immune response triggered by normally harmless environmental substances known as allergens (such as pollen, dust mites, animal dander, fungal spores, or certain foods).

**Immunological Mechanism** The development of an allergic reaction occurs in two distinct phases:

- **1. Sensitization Phase (Initial Exposure):**
  1. An allergen enters the body and is processed by antigen-presenting cells (APCs), which present it to  $CD4^+$  T\_H2 helper cells.
  2. T\_H2 cells secrete cytokines (IL-4, IL-13) that signal B lymphocytes to switch antibody class production to **Immunoglobulin E (IgE)**.
  3. Secreted allergen-specific IgE antibodies bind via their Fc region to high-affinity Fc $\epsilon$  receptors on the surface of tissue mast cells and blood basophils, sensitizing these cells.
- **2. Effector Phase (Re-exposure):**
  1. Upon subsequent exposure, the allergen cross-links adjacent surface-bound IgE molecules on sensitized mast cells.
  2. Cross-linking triggers rapid intracellular signaling leading to **degranulation**, releasing pre-formed chemical mediators including **histamine**, leukotrienes, prostaglandins, and heparin into surrounding tissue.

**Pathological Effects and Symptoms** Released mediators cause smooth muscle contraction, increased vascular permeability, vasodilation, and mucus hypersecretion. Clinical manifestations range from mild local symptoms (rhinitis, hives, conjunctivitis) to severe systemic responses, such as **anaphylaxis**—a life-threatening condition characterized by airway constriction and a rapid drop in blood pressure.

## 24. cDNA Synthesis

**Definition and Concept** Complementary DNA (cDNA) synthesis is the *in vitro* enzymatic process of generating a single-stranded, and subsequently double-stranded, DNA molecule using a mature messenger RNA (mRNA) template. Because eukaryotic pre-mRNA undergoes splicing to remove non-coding introns, cDNA represents only the expressed, coding regions (exons) of a genome.

#### Step-by-Step Enzymatic Procedure

1. **Isolation of mRNA:** Total cellular RNA is extracted, and eukaryotic mRNA is purified using oligo(dT) affinity chromatography, which binds the poly-A tail found on mature eukaryotic transcripts.
2. **Annealing the Primer:** A synthetic primer is annealed to the mRNA template:
  - **Oligo(dT) primer:** Anneals to the 3' poly-A tail.
  - **Random hexamers:** Bind throughout mRNA templates.
  - **Gene-specific primers:** Target a known RNA sequence.
3. **First-Strand cDNA Synthesis:** The enzyme **reverse transcriptase** adds dNTPs to synthesize a complementary DNA strand in the 5' → 3' direction, producing an RNA-DNA hybrid.
4. **RNA Template Removal:** RNase H partially degrades the original RNA strand within the hybrid, leaving small RNA fragments that act as primers, or the RNA strand is completely digested.
5. **Second-Strand cDNA Synthesis:** **DNA Polymerase I** uses the single-stranded cDNA as a template to synthesize the second complementary DNA strand. **DNA Ligase** seals any remaining single-stranded nicks.

**Applications** cDNA synthesis is fundamental for constructing cDNA expression libraries, cloning eukaryotic genes into bacterial plasmids (as bacteria cannot splice introns), and performing quantitative gene expression studies via RT-PCR.

## 25. Construction of Recombinant DNA

**Definition and Concept** The construction of Recombinant DNA (rDNA) involves combining genetic material from two distinct biological sources *in vitro* to create a novel DNA molecule that does not exist in nature. This process forms the foundation of modern gene cloning and genetic engineering.

#### Core Steps in Construction

1. **Isolation of Target DNA and Vector:**
  - The gene of interest (foreign/passenger DNA) is isolated from donor genomic DNA or cDNA.
  - A suitable cloning vector (such as a plasmid or bacteriophage) is isolated from host cells.
2. **Cleavage with Restriction Endonucleases:** Both the insert DNA and the vector DNA are digested using the exact same Type II restriction enzyme (e.g., *EcoRI*). This generates matching, complementary single-stranded overhangs (sticky ends) on both molecules.
3. **Annealing (Ligation):** The digested insert DNA and vector DNA are mixed together. Complementary sticky ends pair via hydrogen bonding between matching base pairs.
4. **Covalent Join by DNA Ligase:** The enzyme **T4 DNA Ligase** is added along with ATP to catalyze the formation of covalent phosphodiester bonds between adjacent 3'-OH and 5'-PO<sub>4</sub> ends, covalently linking the insert into the vector backbone.

5. **Transformation:** The resulting recombinant DNA construct is introduced into an appropriate biological host cell (such as *E. coli*) via transformation for replication and downstream gene expression.

## 26. Shuttle Vectors

**Definition and Key Features** A shuttle vector is a specialized type of recombinant plasmid construct engineered to replicate autonomously in two or more completely different host species (for example, in a prokaryote like *Escherichia coli* and a eukaryote such as *Saccharomyces cerevisiae* or mammalian cells).

**Essential Structural Elements** To function effectively across different biological domains, a shuttle vector must carry dual sets of functional genetic elements:

- **Two Origins of Replication (ori):** A prokaryotic origin of replication (e.g., *pUC ori*) allowing high-copy propagation in *E. coli*, and a eukaryotic origin of replication (e.g., 2 $\mu$  circle origin for yeast, or SV40 origin for mammalian systems).
- **Two Selection Markers:** A bacterial antibiotic resistance gene (such as ampicillin or kanamycin resistance) for selection in bacteria, alongside a eukaryotic selection marker (such as the *URA3* gene for yeast auxotrophs, or neomycin/hygromycin resistance for mammalian cells).
- **Multiple Cloning Site (MCS):** A concentrated region containing unique restriction endonuclease recognition sites for inserting target genes.

**Applications and Advantages** Shuttle vectors allow researchers to perform initial genetic manipulations, plasmid expansions, and sequence verification rapidly within easily grown bacterial hosts, before transferring the intact vector into eukaryotic systems for functional gene expression analysis, protein-protein interaction studies, or functional complementation assays.

## 27. Insertional Inactivation

**Definition and Concept** Insertional inactivation is an *in vitro* selection technique used to identify host bacterial cells that have successfully taken up recombinant plasmids containing an inserted target gene. The method relies on disrupting a functional gene marker within the plasmid vector when foreign DNA is ligated into its sequence.

**Mechanism and Principles** Cloning vectors are deliberately designed so that their Multiple Cloning Site (MCS) lies within the coding sequence of a marker gene.

- **Without Insert (Non-Recombinant Vector):** The marker gene remains intact and produces a functional enzyme or protein.
- **With Insert (Recombinant Vector):** Interruption of the marker gene's coding sequence by inserted foreign DNA disrupts its reading frame, preventing expression of a functional protein.

### Classic Examples

- **1. Disruption of Antibiotic Resistance Genes (e.g., pBR322):**
  - Plasmid pBR322 carries both ampicillin resistance ( $\text{amp}^R$ ) and tetracycline resistance ( $\text{tet}^R$ ) genes.
  - Inserting foreign DNA at the *Bam*HI site inside the  $\text{tet}^R$  gene inactivates tetracycline resistance.
  - Recombinant cells become ampicillin-resistant but tetracycline-sensitive ( $\text{amp}^R\text{tet}^S$ ), allowing identification via replica plating.

- **2. Blue-White Screening (e.g., pUC vectors):**
  - The MCS lies within the *lacZ*α gene, which encodes the α-peptide of β-galactosidase.
  - Non-recombinant vectors produce functional β-galactosidase, which cleaves the chromogenic substrate X-Gal to turn bacterial colonies **blue**.
  - Recombinant vectors undergo insertional inactivation of *lacZ*α, failing to produce active enzyme and resulting in **white** colonies.

## 28. Gene Cloning

**Definition and Fundamental Principle** Gene cloning is the molecular process of isolating a specific gene or DNA sequence from a donor organism and inserting it into a replicative vector to generate multiple identical copies within a host cell.

### Core Steps in Gene Cloning

1. **Isolation of Target DNA:** The specific gene sequence is isolated using restriction digestion, cDNA synthesis, or PCR amplification.
2. **Selection of Vector:** A suitable cloning vehicle (plasmid, bacteriophage, cosmid, or BAC) is chosen based on insert size requirements.
3. **Ligation:** Target DNA and vector DNA are cleaved with the same restriction enzyme and covalently joined using DNA ligase to form recombinant DNA.
4. **Transformation:** The recombinant vector is introduced into competent host cells (e.g., *E. coli*) using heat shock, electroporation, or chemical treatments.
5. **Selection and Screening:** Host cells are grown on selective media (containing antibiotics) and screened (via blue-white selection or colony PCR) to identify transformants carrying recombinant plasmids.
6. **Multiplication and Expression:** Selected host clones are cultured to produce millions of gene copies or harvested for downstream protein expression.

**Applications** Gene cloning is widely used to produce recombinant proteins (such as insulin), map genomes, generate transgenic plants and animals, and analyze gene structure and function.

## 29. Restriction Mapping

**Definition and Concept** Restriction mapping is an analytical laboratory method used to map the relative locations of restriction endonuclease cleavage sites along a linear or circular piece of DNA. The resulting map acts as a structural blueprint of the DNA fragment.

### Experimental Methodology

1. **Digestion Reactions:** Aliquots of purified target DNA are subjected to individual and combined single/double digests using specific restriction enzymes:
  - Reaction 1: Enzyme A alone.
  - Reaction 2: Enzyme B alone.
  - Reaction 3: Enzyme A + Enzyme B simultaneously (double digest).
2. **Gel Electrophoresis:** The resulting restriction fragments from each reaction are separated by size on an agarose gel alongside DNA molecular weight markers.
3. **Fragment Size Analysis:** Fragment sizes are calculated based on migration distances.
4. **Map Construction:** By comparing fragment sizes generated by single digests against double digests, researchers deduce the order and physical distances (in base pairs or

kilobases) between restriction sites.

**Applications** Restriction mapping provides a fundamental tool for verifying recombinant plasmid constructs, confirming gene insertion orientation, detecting structural genomic rearrangements, and analyzing DNA fragments prior to full nucleotide sequencing.

## 30. Repetitive DNA Sequences (*Duplicate question answered with focus on organizational patterns*)

**Structural Organization and Chromosomal Distribution** Repetitive DNA comprises non-coding nucleotide sequences present in multiple copies throughout eukaryotic genomes. Structurally, these repetitions are organized into two primary patterns:

- **Tandem Repetitions:** Short or long units linked end-to-end in direct repetition, localized predominantly at heterochromatic regions such as centromeres and telomeres. These include:
  - **Satellite DNA:** Large, highly repetitive clusters spanning up to megabases.
  - **Minisatellites (VNTRs):** Medium-sized repeat units (10–100 bp) used in population genetics and forensic science.
  - **Microsatellites (STRs):** Very short repeats (2–6 bp) distributed throughout euchromatin, useful for genetic mapping due to high allele diversity.
- **Interspersed Repetitions:** Individual repetitive elements dispersed throughout euchromatic regions, largely derived from mobile retrotransposons:
  - **LINEs (Long Interspersed Elements):** Autonomously transposing units encoding reverse transcriptase enzymes.
  - **SINEs (Short Interspersed Elements):** Non-autonomous units reliant on LINE machinery for movement.

**Functional Relevance** Although historically referred to as "junk DNA," repetitive sequences play essential roles in maintaining structural chromosome integrity, facilitating spindle attachment during cell division, driving evolutionary novelty via recombination, and regulating non-coding RNA production.

## 31. Satellite DNA

**Definition and Density Gradient Behavior** Satellite DNA represents a distinct class of highly repetitive, non-coding tandem DNA sequences concentrated around centromeric and telomeric chromosomal regions. The term "satellite" originates from cesium chloride ( $\text{CsCl}$ ) equilibrium density gradient centrifugation experiments, where these repetitive sequences band separately as minor "satellite" peaks apart from the main genomic DNA peak due to differences in base composition (GC vs. AT richness).

**Sub-Classes Based on Repeat Unit Size** Satellite DNA is broadly classified based on the length of its repeating unit and total array size:

- **Macro-satellite DNA:** Large arrays spanning hundreds of kilobases with repeat units ranging from 100 to over 1000 base pairs (e.g.,  $\alpha$ -satellite DNA at human centromeres, based on ~170 bp repeat units).
- **Minisatellite DNA (VNTRs):** Intermediate repeat units ranging from 10 to 100 base pairs, spanning total array lengths of 0.5 to 20 kb.
- **Microsatellite DNA (STRs):** Short tandem repeats of 2 to 6 base pairs (such as

[ $\text{CA}$ ]<sub>n</sub> repeats), spanning array lengths under 100 bp.

### Biological Functions

- **Centromere Structure and Function:** Centromeric satellite arrays serve as binding sites for kinetochore proteins required for proper spindle attachment and chromosome segregation during cell division.
- **Heterochromatin Formation:** Contributes to heterochromatin condensation and gene silencing.
- **Genetic Profiling:** High length polymorphism in mini- and microsatellites forms the foundation of modern DNA fingerprinting and linkage mapping.

## 32. Plasmid as Cloning Vector

**Definition and General Features** Plasmids are naturally occurring, extra-chromosomal, double-stranded, circular DNA molecules found predominantly in bacteria that replicate autonomously and independently of the host genomic chromosome. Natural plasmids often carry non-essential genes that confer evolutionary advantages, such as antibiotic resistance. Artificially modified plasmids serve as core cloning vectors in molecular biology.

### Essential Features of an Ideal Plasmid Vector

1. **Origin of Replication (*ori*):** A specific DNA sequence required for initiating autonomous replication inside the host cell, controlling plasmid copy number.
2. **Selectable Marker Genes:** Genes encoding resistance to antibiotics (e.g.,  $\text{amp}^R$ ,  $\text{tet}^R$ ,  $\text{kan}^R$ ) that allow selective growth of transformed bacteria on media containing the corresponding drug.
3. **Multiple Cloning Site (MCS / Polylinker):** A artificially engineered sequence containing unique recognition sites for multiple restriction endonucleases, allowing target DNA fragments to be inserted without disrupting other essential plasmid functions.
4. **Small Size:** Ideally small (typically 3 to 6 kb) to ensure easy isolation, manipulation, and efficient uptake during bacterial transformation.

### Classic Examples

- **pBR322:** Early standard cloning vector carrying ampicillin and tetracycline resistance markers.
- **pUC Series (e.g., pUC19):** High-copy-number vectors equipped with an MCS located within the *lacZ*α gene, enabling rapid blue-white screening.

## 33. Structure of Immunoglobulin

**Definition and Basic Architecture** Immunoglobulins (Igs), or antibodies, are antigen-binding glycoprotein molecules synthesized by plasma B lymphocytes in response to foreign immunogens. A standard immunoglobulin monomer possesses a characteristic symmetrical Y-shaped structure built from four polypeptide chains held together by covalent interchain disulfide bonds and non-covalent interactions.

**Chain Composition** A typical antibody monomer (150 kDa) consists of:

- **Two Identical Heavy (H) Chains:** Larger polypeptide chains (~50 kDa each).
- **Two Identical Light (L) Chains:** Smaller polypeptide chains (~25 kDa each). Light chains occur in two forms: kappa ( $\kappa$ ) or lambda ( $\lambda$ ).

### Structural Domains and Functional Regions

- **Variable Regions ( $V_H$  and  $V_L$ ):** Located at the amino-terminal (N-terminal) ends of

both heavy and light chains. These regions vary in amino acid sequence between antibodies and contain hypervariable regions termed **Complementarity-Determining Regions (CDRs)** that form the specific antigen-binding site (paratope).

- **Constant Regions (C<sub>H</sub> and C<sub>L</sub>):** Located toward the carboxyl-terminal (C-terminal) ends. The heavy chain constant region determines the antibody's immunoglobulin class ( $\text{IgG}$ ,  $\text{IgM}$ ,  $\text{IgA}$ ,  $\text{IgD}$ , or  $\text{IgE}$ ) and mediates effector functions such as complement activation and binding to cell surface Fc receptors.

#### **Proteolytic Cleavage Products**

- **Papain Digestion:** Cleaves the heavy chains above the disulfide hinge region, yielding two identical **Fab** (Fragment antigen-binding) fragments and one **Fc** (Fragment crystallizable) fragment.
- **Pepsin Digestion:** Cleaves below the hinge region, producing a single divalent  $\text{F(ab')}_2$  fragment and digested heavy chain fragments.

**Question - Discuss the role of genetic engineering in agriculture, animal husbandry, and pollution control.**

**Answer** - Genetic engineering, or recombinant DNA technology, involves the direct manipulation of an organism's genome using biotechnology. Over the last few decades, it has revolutionized applied biological sciences, particularly agriculture, animal breeding, and environmental management.

**Role in Agriculture:** Genetic engineering has enabled the creation of Genetically Modified (GM) crops designed for higher yields, biotic stress resistance, and abiotic stress tolerance.

1. *Insect and Pest Resistance:* The most prominent example is Bt cotton and Bt brinjal, engineered using genes from the soil bacterium *Bacillus thuringiensis*. These genes encode crystal proteins ( $\delta$ -endotoxins) that kill specific lepidopteran and coleopteran pests upon ingestion, dramatically reducing chemical pesticide usage.
2. *Herbicide Tolerance:* Crops like glyphosate-tolerant Roundup Ready soybeans allow farmers to spray broad-spectrum herbicides without harming the crop plants.
3. *Nutritional Enhancement:* Biofortification through genetic engineering addresses global micronutrient deficiencies. "Golden Rice," enriched with  $\beta$ -carotene (precursor of Vitamin A) using genes from daffodils and *Erwinia*, serves as a prime example.
4. *Abiotic Stress Tolerance:* Genes regulating osmoregulation and heat shock proteins are transferred into staple crops like wheat and rice to survive drought, salinity, and extreme temperatures.

**Role in Animal Husbandry:** Transgenic technology in livestock focuses on enhancing productivity, disease resistance, and bio-farming.

1. *Enhanced Milk Production and Quality:* Transgenic cows and goats have been developed to produce milk with higher protein content or human proteins like  $\alpha$ -1-antitrypsin and human lactoferrin (e.g., "Rosie" the transgenic cow).
2. *Disease Resistance:* Genetic modifications aim to render livestock resistant to endemic infections, such as bovine spongiform encephalopathy (BSE) in cattle and mastitis in dairy cows.
3. *Growth Rate Acceleration:* Transgenic pigs and salmon engineered with additional copies of growth hormone (GH) genes exhibit significantly faster growth rates and better feed conversion efficiency.

**Role in Pollution Control (Bioremediation):** Biotechnology plays a crucial role in cleaning contaminated natural environments using genetically engineered microorganisms (GEMs).

1. *Hydrocarbon Degradation:* Ananda Mohan Chakrabarty engineered the famous "Superbug" (*Pseudomonas putida*), which harbors plasmids encoding enzymes that rapidly break down various components of crude oil slicks.
2. *Heavy Metal Remediation:* Recombinant strains of *Escherichia coli* and *Deinococcus radiodurans* express metallothionein proteins and phytochelatins, allowing them to absorb and isolate toxic heavy metals like mercury, cadmium, and arsenic from industrial effluents.
3. *Xenobiotic Degradation:* Microbes are engineered with catabolic pathways to break down non-biodegradable synthetic pollutants, such as polychlorinated biphenyls (PCBs), pesticides (like DDT), and plastics.

**Question - What are restriction enzymes? Differentiate between restriction endonuclease type I, II, and III. Mention the uses of these enzymes in gene cloning.**

**Answer** - Restriction enzymes (restriction endonucleases) are bacterial enzymes that cleave double-stranded DNA at specific nucleotide sequences known as recognition sites or

palindromic sequences. They act as a natural defense mechanism (restriction-modification system) in bacteria against invading bacteriophages by digesting viral DNA.

### **Classification and Comparison of Restriction Endonucleases:**

#### **1. Type I Restriction Endonucleases:**

- **Structure:** Complex multifunctional enzymes containing three different subunits (R for restriction, M for modification, S for specificity).
- **Requirements:** Require ATP, S-adenosylmethionine (SAM), and  $Mg^{2+}$  ions as cofactors.
- **Cleavage Site:** They recognize a specific sequence but cut randomly at a distance far away (often 1000 base pairs or more) from the recognition site.
- **Enzyme Activity:** Possess both endonuclease and methylase activities in the same complex.
- **Cloning Utility:** Unsuitable for molecular cloning due to variable, non-specific cut sites.

#### **2. Type II Restriction Endonucleases:**

- **Structure:** Simpler homodimers that act as separate endonuclease and methylase complexes.
- **Requirements:** Require only  $Mg^{2+}$  ions as cofactors.
- **Cleavage Site:** Cleave DNA precisely within or at fixed, short distances (usually 4 to 8 base pair palindromic sequences) from their recognition sites.
- **Enzyme Activity:** Endonuclease and methylase activities are completely independent.
- **Cloning Utility:** Highly suitable and indispensable for recombinant DNA technology and gene cloning (e.g., *EcoRI*, *HindIII*, *BamHI*).

#### **3. Type III Restriction Endonucleases:**

- **Structure:** Heterodimeric enzymes composed of two different subunits (Res and Mod).
- **Requirements:** Require ATP and  $Mg^{2+}$  ions; SAM acts as a stimulant.
- **Cleavage Site:** Cleave DNA outside the recognition sequence at a specific short distance (around 24 to 26 base pairs downstream).
- **Enzyme Activity:** Endonuclease and methylase activities are present in a single complex.
- **Cloning Utility:** Rarely used in recombinant DNA work due to incomplete digestion and complex cut mechanics.

### **Uses of Restriction Enzymes in Gene Cloning:**

1. **Construction of Recombinant DNA:** Type II enzymes cut both foreign passenger DNA and vector DNA (like pBR322 or pUC plasmids) at identical palindromic sites, generating compatible cohesive ("sticky") ends or blunt ends. These ends are easily ligated using DNA ligase.
2. **Gene Mapping:** They are used in restriction fragment length polymorphism (RFLP) and physical mapping of genomes.
3. **Diagnostic Cleavage:** To verify whether a plasmid vector has successfully incorporated an insert gene by performing restriction digest analysis.

### **Question - Discuss the potential hazards of recombinant DNA technology.**

**Answer -** Recombinant DNA (rDNA) technology has transformed genetics, medicine, and agriculture. However, the creation of novel genetic combinations and transgenic organisms poses biosafety, ethical, and environmental concerns.

### **Environmental and Ecological Hazards:**

1. *Gene Flow and Superweeds*: Transgenes conferring herbicide resistance can escape from genetically modified crops to wild relatives through cross-pollination. This may produce invasive "superweeds" that cannot be controlled by conventional herbicides.
2. *Impact on Non-Target Species*: Bt crops express toxins that kill harmful caterpillars, but residues in pollen may harm non-target beneficial insects, such as monarch butterfly larvae or essential soil microbes.
3. *Loss of Biodiversity*: Widespread monoculture of uniform GM crops displaces traditional landraces, leading to genetic erosion and ecosystem fragility.

#### **Human Health Hazards:**

1. *Allergenicity and Toxicity*: Introducing novel proteins into food crops can provoke unpredictable allergic reactions in human populations (e.g., the introduction of a Brazil nut allergen into GM soybeans).
2. *Antibiotic Resistance Spread*: Selectable marker genes used during recombinant selection frequently encode resistance to antibiotics (e.g., ampicillin, kanamycin). Horizontal gene transfer from transgenic food to gut bacteria could spread antibiotic resistance.
3. *Carcinogenesis or Oncogenesis*: Gene therapy vectors using viral systems (like retroviruses) risk random integration into the host genome, which can accidentally activate proto-oncogenes and trigger leukemia.

#### **Biological Safety and Bio-warfare Risks:**

1. *Escape of Genetically Engineered Microorganisms (GEMs)*: Modified viruses, bacteria, or fungi escaping research facilities could upset natural microbial balances or cause novel, untreatable human or animal diseases.
2. *Bioweapons Development*: The ability to alter pathogens creates risks that malicious entities could engineer drug-resistant, high-mortality strains of viruses (e.g., modified poxviruses or anthrax bacteria) for biological warfare.

**Socio-Ethical Concerns:** Patenting GM seeds and life forms grants multinational corporations monopolies over essential agricultural resources, threatening seed sovereignty for small farmers in developing nations.

#### **Question - Describe the various cellular components of the immune system.**

**Answer -** The immune system comprises specialized cells, tissues, and organs working collectively to defend the host against pathogens. The cellular elements originate from pluripotent hematopoietic stem cells in the bone marrow and differentiate into myeloid and lymphoid lineages.

#### **Lymphocytes (Adaptive Immunity):**

1. *T-Lymphocytes (T-Cells)*:
  - Originating in the bone marrow and maturing in the thymus, T-cells mediate cell-mediated immunity (CMI).
  - *Helper T-cells (CD4<sup>+</sup>)*: Recognize antigens presented on MHC Class II molecules and secrete cytokines (IL-2, IL-4, IFN-gamma) that activate B-cells, cytotoxic T-cells, and macrophages.
  - *Cytotoxic T-cells (CD8<sup>+</sup>)*: Recognize endogenous antigens on MHC Class I molecules and directly destroy virus-infected or tumor cells by releasing perforins and granzymes.
  - *Regulatory T-cells (T<sub>reg</sub>)*: Suppress autoreactive immune responses and maintain immune tolerance.
2. *B-Lymphocytes (B-Cells)*:
  - Maturing in the bone marrow (or Bursa of Fabricius in birds), B-cells drive humoral

immunity.

- Upon encountering specific antigens, they proliferate into short-lived *Plasma Cells*, which secrete large amounts of antibodies (immunoglobulins like IgG, IgM, IgA), and long-lived *Memory B-cells*, which ensure a rapid secondary response.
3. **Natural Killer (NK) Cells:**
- Innate lymphoid cells that target tumor cells and virus-infected host cells without prior sensitization. They recognize down-regulated MHC Class I molecules and kill target cells via antibody-dependent cellular cytotoxicity (ADCC).

### **Phagocytic Cells (Innate Immunity):**

1. **Neutrophils:** The most abundant white blood cells (60-70%). They act as first responders to tissue inflammation, migrating via chemotaxis to ingest pathogens through phagocytosis and destroy them using reactive oxygen species (ROS) and lysosomal enzymes.
2. **Macrophages and Monocytes:** Monocytes circulate in the blood before migrating into tissues to become macrophages (e.g., Kupffer cells in the liver, Microglia in the brain). They serve dual roles: engulfing cellular debris and acting as professional Antigen Presenting Cells (APCs).
3. **Dendritic Cells:** Star-shaped cells residing in barrier tissues. They act as the primary bridge between innate and adaptive immunity by capturing antigens and displaying them on MHC molecules to naive T-cells in lymph nodes.

### **Granulocytes Involved in Hypersensitivity and Defense:**

1. **Eosinophils:** Protect against large multicellular parasites (e.g., helminths) by releasing major basic protein (MBP) and play a key role in allergic reactions.
2. **Basophils and Mast Cells:** Rich in histamine, heparin, and serotonin granules. They express high-affinity Fc receptors for IgE antibodies, triggering degranulation during allergic responses and anaphylaxis.

### **Question - What is DNA fingerprinting? Discuss its applications.**

**Answer -** DNA fingerprinting, also known as DNA profiling or DNA typing, is a laboratory technique used to establish the genetic identity of an individual by analyzing unique variations in their DNA sequences. Developed by Sir Alec Jeffreys in 1984, the method relies on hypervariable region variations called VNTRs (Variable Number of Tandem Repeats) and STRs (Short Tandem Repeats) present within non-coding regions of the genome.

**Principle and Methodology:** While 99.9% of human DNA is identical across individuals, the remaining 0.1% exhibits polymorphic satellite DNA. DNA fingerprinting involves:

1. Extraction of genomic DNA from biological samples (blood, semen, hair follicle, saliva).
2. Cleavage of DNA using specific restriction endonucleases.
3. Separation of fragments by agarose gel electrophoresis.
4. Transfer of separated fragments to a nylon membrane via Southern Blotting.
5. Hybridization using radioactively or fluorescently labeled VNTR probes.
6. Detection of unique band patterns using autoradiography or fluorescence imaging.

### **Applications of DNA Fingerprinting:**

1. **Forensic Investigations:**
  - DNA fingerprinting is widely used in criminal justice. Biological evidence recovered from crime scenes (e.g., bloodstains, semen, or hair roots) is compared against suspect DNA profiles. A match provides definitive evidence linking a suspect to the scene.
2. **Paternity and Maternity Disputes:**
  - Because individuals inherit half their VNTR alleles from the biological mother and

half from the biological father, comparing band patterns reliably resolves parentage and inheritance disputes.

3. *Pedigree Analysis and Wildlife Conservation:*

- Used by conservation biologists to assess genetic diversity, prevent inbreeding depression in endangered species populations, and curb illegal poaching (e.g., identifying ivory sources or illegal trade in animal derivatives).

4. *Taxonomy and Phylogenetics:*

- Helps establish evolutionary relationships, taxonomical classifications, and evolutionary links among species.

5. *Medical Diagnostics and Bone Marrow Transplants:*

- Used to monitor engraftment success after bone marrow transplantation by evaluating host versus donor cell populations, and to detect specific genetic disorders or somatic mutations.

**Question - Define transposons. Provide a detailed account of composite and non-composite transposons and their importance in prokaryotes.**

**Answer -** Transposons, or Transposable Elements (TEs), are mobile DNA sequences capable of moving ("jumping") from one location in a genome to another. Discovered by Barbara McClintock in maize, these "jumping genes" occur in both prokaryotic and eukaryotic organisms.

**Composite vs. Non-Composite Transposons:**

1. *Composite Transposons (Tn1 to Tn10 family):*

- *Structure:* These transposable elements consist of a central, non-transposon gene region (frequently carrying antibiotic resistance genes like tetracycline or kanamycin resistance) flanked on both ends by Insertion Sequence (IS) elements.
- *Mechanism:* The flanking IS elements provide the necessary transposase enzyme and inverted repeat termini. The IS elements can move independently or carry the entire central module along to a new target site.
- *Example:* Tn10, which carries a tetracycline resistance gene flanked by inverted IS10 elements.

2. *Non-Composite Transposons (Tn3 family):*

- *Structure:* Non-composite transposons do not possess IS elements at their ends. Instead, they feature short inverted repeat sequences (20 to 40 base pairs) at both termini.
- *Gene Content:* The central segment encodes its own transposition enzymes—specifically Transposase (tnpA) and Resolvase (tnpR)—alongside accessory genes such as  $\beta$ -lactamase (conferring ampicillin resistance).
- *Mechanism:* They move primarily via a replicative transposition process, forming a co-integrate intermediate that is later resolved.
- *Example:* Tn3, which carries resistance to ampicillin.

**Importance of Transposons in Prokaryotes:**

1. *Spread of Multi-Drug Resistance:* Transposons move between plasmids (such as R-plasmids) and the bacterial chromosome. Conjugative plasmids can then transfer these resistance transposons across different species, driving multidrug-resistant "superbug" strains.
2. *Induction of Mutations:* When a transposon inserts into a functional coding region or regulatory element, it disrupts the reading frame, causing insertion mutations or altered gene expression.
3. *Genome Plasticity and Evolution:* Transposons promote chromosomal rearrangements, including duplications, deletions, and inversions, serving as a primary driver of bacterial

genome evolution.

**Question - Discuss the various types of repetitive DNA sequences.**

**Answer** - Repetitive DNA consists of nucleotide sequences present in multiple copies throughout a genome. While prokaryotic genomes consist largely of unique coding sequences, eukaryotic genomes contain substantial amounts of non-coding repetitive DNA (often termed "junk DNA").

**Major Types of Repetitive DNA Sequences:**

1. *Tandemly Repetitive DNA (Satellite DNA):*
  - These sequences sit head-to-tail in arrays located around centromeric and telomeric regions. They are categorized by unit length:
  - *Macro-Satellites (Classic Satellite DNA):* Long blocks of tandem repeats (ranging from 100 to several hundred base pairs) concentrated near centromeres.
  - *Mini-Satellites (VNTRs - Variable Number of Tandem Repeats):* Units of 10 to 60 base pairs repeated up to several hundred times. They show high individual variability, forming the basis of DNA fingerprinting.
  - *Micro-Satellites (STRs - Short Tandem Repeats / SSRs):* Short sequences of 1 to 6 base pairs (e.g., (CA)<sub>n</sub> repeats) repeated 10 to 100 times. They are widely used as genetic markers in mapping and forensically.
2. *Interspersed Repetitive DNA (Transposable Element-Derived):*
  - These elements are dispersed throughout the genome rather than arranged tandemly, mainly originating from retrotransposons:
  - *SINEs (Short Interspersed Nuclear Elements):* Non-autonomous retrotransposons 100 to 500 base pairs long that rely on external enzymes for movement. The most prominent example in primates is the *Alu* element.
  - *LINEs (Long Interspersed Nuclear Elements):* Autonomous retrotransposons 6,000 to 8,000 base pairs long that encode reverse transcriptase and endonuclease enzymes (e.g., LINE-1 or L1 in humans).
3. *Gene Families (Coding Repetitive DNA):*
  - Essential sequences present in multiple functional copies to meet high cellular demand, such as Histone gene clusters, Ribosomal RNA (rRNA) genes, and Transfer RNA (tRNA) genes.

**Question - Describe how the genome is organized in eukaryotes.**

**Answer** - Eukaryotic genomes are considerably larger and more complex than prokaryotic genomes. A human cell contains roughly 2 meters of linear DNA packed inside a nucleus measuring only 5 to 10 micrometers in diameter. Achieving this degree of compaction requires a structured hierarchy of chromatin folding.

**Packaging Levels of the Eukaryotic Genome:**

1. *The Nucleosome Core ("Beads-on-a-String" 10 nm Fiber):*
  - The nucleosome represents the fundamental repeating unit of eukaryotic chromatin.
  - An octamer composed of basic histone proteins (two molecules each of H2A, H2B, H3, and H4) forms a core around which approximately 146 base pairs of negatively charged DNA wrap 1.65 times.
  - Nucleosomes are joined by linker DNA (20 to 80 base pairs) associated with Histone H1. This structural organization reduces DNA length roughly 6-fold.
2. *Solenoid / Chromatin Fiber (30 nm Fiber):*
  - Histone H1 drives the coiling of the 10 nm nucleosome chain into a helical solenoid structure measuring roughly 30 nm in diameter. This stage provides an additional 40-fold compaction.

3. *Looped Domains (300 nm) and Scaffold Attachment:*
  - The 30 nm fiber forms structural loops attached to a non-histone protein scaffold (nuclear matrix) through Matrix Attachment Regions (MARs) or Scaffold Attachment Regions (SARs).
4. *Higher-Order Coiling and Mitotic Chromosome:*
  - During cell division (metaphase), looped domains condense further with condensin and topoisomerase II, yielding the fully condensed mitotic chromosome, achieving a total compaction ratio of approximately 10,000-fold.

#### Functional Genome Segmentation:

- *Euchromatin:* Less condensed, lightly staining, transcriptionally active regions.
- *Heterochromatin:* Highly condensed, darkly staining, transcriptionally inactive regions (divided into *constitutive*, like centromeres, and *facultative*, like the inactivated female X-chromosome/Barr body).

**Question - What are intervening sequences in eukaryotic genes? How are they removed? Write a note on their significance.**

**Answer -** Unlike prokaryotic genes, eukaryotic protein-coding genes are "split genes." Intervening sequences, termed **Introns**, are non-coding segments interspersed between coding regions known as **Exons**. Introns are transcribed into pre-mRNA but are removed before translation.

**Mechanism of Intron Removal (RNA Splicing):** Introns are excised from pre-mRNA in the nucleus by a macromolecular machine called the **Spliceosome**, composed of small nuclear ribonucleoproteins (snRNPs: U1, U2, U4, U5, U6) and auxiliary proteins.

1. *Recognition of Splicing Signals:*
  - The 5' end of the intron contains a conserved 5'-GU} donor site, the 3' end contains a conserved AG-3' acceptor site, and an internal Adenine residue serves as the Branch Point Site.
2. *First Transesterification Reaction:*
  - The U1 snRNP binds to the 5' splice site, and U2 binds to the branch point.
  - The 2'-OH} group of the branch point Adenine attacks the 5' phosphate at the intron donor site, cleaving the 5' exon-intron boundary and forming a branched "lariat" intermediate structure.
3. *Second Transesterification Reaction:*
  - The free 3'-OH} group of the upstream exon attacks the 5' phosphate of the downstream exon at the 3' acceptor site.
  - This joins the exons together with a phosphodiester bond and releases the excised intron as a lariat, which is degraded.

#### Significance of Intervening Sequences:

1. *Alternative Splicing:* A single pre-mRNA transcript can be spliced in multiple ways by combining different exons. This mechanism allows one gene to encode multiple distinct protein isoforms, expanding proteomic diversity.
2. *Exon Shuffling and Evolution:* Introns provide recombination hotspots where exons can be duplicated or swapped between non-homologous genes over evolutionary time, generating novel functional proteins.
3. *Gene Expression Regulation:* Intron sequences often harbour regulatory elements, such as enhancers and microRNAs, that influence transcriptional efficiency and mRNA transport.

**Question - How can a fragment of DNA be introduced into a bacterial cell? Discuss the strategy by which introduction of the fragment will be confirmed.**

**Answer** - Introducing a foreign DNA fragment into a host bacterial cell (gene transfer) is a critical step in genetic engineering. Because the bacterial cell wall and cell membrane carry net negative charges that repel negatively charged DNA molecules, specialized methods are needed to facilitate uptake.

#### **Methods of Introducing DNA Fragments into Bacteria:**

1. *Chemical Transformation:*
  - Bacterial cells (e.g., *E. coli*) are treated with ice-cold calcium chloride (CaCl<sub>2</sub>). Ca<sup>2+</sup> ions neutralize the negative charges on both the DNA backbone and cell membrane phosphates.
  - A brief heat shock (42°C for 45–90 seconds) followed by rapid chilling on ice alters membrane fluidity, creating temporary pores through which recombinant DNA enters.
2. *Electroporation:*
  - Host bacterial cells are subjected to a high-voltage electrical pulse. This transiently destabilizes the plasma membrane, inducing temporary hydrophobic pores through which DNA enters.
3. *Bacteriophage Transduction / In Vitro Packaging:*
  - Foreign DNA is ligated into viral vectors (e.g., Lambda phage or cosmids) and packaged into viral protein capsids *in vitro*. These phages then infect target bacteria naturally, injecting recombinant DNA.

#### **Strategies for Confirming and Selecting Transformants:**

1. *Antibiotic Resistance Marker Selection:*
  - Cloning vectors carry antibiotic resistance genes (e.g., amp<sup>R</sup> for ampicillin resistance). Cells are plated on selective media containing ampicillin; only bacteria that successfully take up the plasmid grow into colonies.
2. *Insertional Inactivation (Blue-White Screening):*
  - Vectors like pUC19 carry the lacZ $\alpha$  gene coding for the beta-galactosidase enzyme. The multiple cloning site (MCS) lies within lacZ $\alpha$ .
  - *Non-Recombinant Plasmids (Vector without insert):* Intact lacZ $\alpha$  produces functional beta-galactosidase, which cleaves the substrate X-Gal on agar plates to yield **blue colonies**.
  - *Recombinant Plasmids (Vector with insert):* The inserted DNA disrupts lacZ $\alpha$  (insertional inactivation), preventing beta-galactosidase synthesis and resulting in **white colonies**.
3. *Colony PCR and Restriction Digest Verification:*
  - Primers flanking the insertion site amplify the target insert from isolated white colonies via PCR. Alternatively, plasmid DNA is extracted, digested with restriction enzymes, and run on agarose gel to confirm the fragment size.

**Question - What is somatic hybridization? Elucidate this process in case of malignancy and mention its applications in animals.**

**Answer** - Somatic hybridization (somatic cell fusion) is a non-sexual technique involving the *in vitro* fusion of somatic cells derived from different strains, tissues, or species to yield a viable hybrid cell containing genomic contributions from both parents.

#### **Process of Somatic Cell Fusion:**

1. *Cell Isolation and Preparation:* Somatic cells are isolated and cultured.
2. *Induction of Fusion:* Cell membranes are brought together using fusogenic agents—most notably Polyethylene Glycol (PEG) or inactivated Sendai virus (Hemagglutinating Virus of Japan)—or through electrofusion.

3. *Heterokaryon Formation*: Membranes fuse to yield a bi-nucleated cell called a heterokaryon.
4. *Synkaryon Formation*: During subsequent mitotic cell divisions, the nuclear membranes dissolve and chromosomes merge into a single hybrid nucleus (synkaryon).

**Somatic Hybridization in Malignancy Research**: Somatic hybridization played a key role in clarifying cancer genetics, particularly through cell fusion experiments conducted by Henry Harris and colleagues:

- When a malignant cancer cell is fused with a normal non-malignant somatic cell, the resulting hybrid cell often displays a non-malignant phenotype.
- This demonstrates that malignancy is typically recessive and that normal somatic cells supply active **Tumor Suppressor Genes** (such as p53 or Rb) that suppress the cancerous phenotype.
- As the somatic hybrid cell divides, it may randomly shed chromosomes. Loss of specific chromosomes carrying these tumor suppressor genes leads to a return of the malignant phenotype, helping researchers map tumor suppressor locations to specific human chromosomes.

#### **Applications in Animal Biotechnology:**

1. *Monoclonal Antibody Production (Hybridoma Technology)*: Developed by Köhler and Milstein, antibody-producing B-lymphocytes are fused with immortal myeloma (bone marrow cancer) cells. The resulting hybridoma cell lines continuously produce monoclonal antibodies (mAbs) for diagnostic and therapeutic uses.
2. *Gene Mapping and Chromosome Localization*: In human-rodent somatic hybrids, human chromosomes are preferentially lost over time. Correlating the presence of specific human enzymes with remaining human chromosomes helped assign genes to specific chromosomes before high-throughput sequencing.
3. *Analysis of Gene Regulation*: Somatic hybrids help researchers study dominant/recessive interactions among cytoplasmic and nuclear factors governing gene expression.

**Question - Give an account of mRNA synthesis in prokaryotes. How does it differ from that of eukaryotes?**

**Answer** - Synthesis of messenger RNA (mRNA) from a DNA template—termed **Transcription**—is catalyzed in prokaryotes by a single DNA-dependent RNA polymerase enzyme.

#### **Steps of Prokaryotic mRNA Synthesis:**

1. *Initiation*:
  - The prokaryotic RNA polymerase holoenzyme (recognizes specific promoter sequences upstream of the transcription start site (+1), specifically the -10 box (Pribnow box: 5'-TATAAT-3' and -35 box 5'-TTGACA-3'.
  - The sigma ( $\sigma$ ) subunit directs the enzyme to the promoter, forming a closed complex. Unwinding of DNA creates an open complex, and RNA synthesis begins. Once an RNA transcript reaches ~10 nucleotides, the sigma factor dissociates.
2. *Elongation*:
  - The core enzyme ( $\alpha_2\beta\beta'\omega$ ) moves along the template strand in the 3' to 5' direction, synthesizing mRNA continuously in the 5' to 3' direction using ribonucleoside triphosphates (ATP, UTP, CTP, GTP).
3. *Termination*:
  - *Rho-Independent (Intrinsic) Termination*: The transcript forms a GC-rich stem-loop (hairpin) structure followed by a run of Uracil (U) residues. This hairpin stalls RNA polymerase, destabilizing the A-U hybrid region and releasing the mRNA.

- *Rho-Dependent Termination*: A protein factor (rho hexamer ATPase) binds to a rut site on the emerging RNA, migrates toward the polymerase, and unwinds the RNA-DNA hybrid.

### Key Differences Between Prokaryotic and Eukaryotic mRNA Synthesis:

1. *RNA Polymerases*: Prokaryotes use a single RNA polymerase for all cellular RNAs. Eukaryotes use three distinct enzymes: RNA Polymerase I (rRNA), RNA Polymerase II (mRNA), and RNA Polymerase III (tRNA and 5S rRNA).
2. *Spatial and Temporal Coupling*: Prokaryotic transcription and translation occur concurrently in the cytoplasm (coupled transcription-translation). Eukaryotic transcription occurs in the nucleus, and translation takes place later in the cytoplasm.
3. *Post-Transcriptional Modifications*: Prokaryotic mRNA is synthesized in mature form without processing. Eukaryotic pre-mRNA undergoes extensive nuclear modifications: 5' capping (7-methylguanosine), 3' polyadenylation (poly-A tail addition), and RNA splicing (removal of introns).
4. *Transcript Architecture*: Prokaryotic mRNA is commonly **polycistronic** (encoding multiple distinct proteins from a single operon). Eukaryotic mRNA is **monocistronic** (encoding a single polypeptide).

### Question - Describe the Maxam and Gilbert procedure of DNA sequencing.

**Answer** - Developed by Allan Maxam and Walter Gilbert in 1977, the Maxam-Gilbert method (also known as the chemical cleavage method) was one of the first widely adopted approaches for determining DNA nucleotide sequences.

**Principle**: The method relies on end-labeling DNA fragments with radioactive isotope markers, followed by base-specific chemical modifications that selectively break the DNA backbone at particular bases (A, G, C, or T).

#### Step-by-Step Procedure:

1. *DNA Labeling and Strand Separation*:
  - Single-stranded or double-stranded target DNA is radiolabeled at its 5' end using [ $\gamma\text{-}^{32}\text{P}$ ] ATP and Polynucleotide Kinase.
  - If double-stranded, strands are separated and isolated.
2. *Aliquoting and Chemical Reactions*:
  - The labeled DNA sample is split into four separate test tubes (G, A+G, C+T, and C).
  - Base-specific chemical reagents are added under controlled conditions to achieve sparse, partial cleavage (averaging one cleavage event per DNA molecule):
  - *Tube G*: Treated with **dimethyl sulfate (DMS)**, which selectively methylates Guanine residues.
  - *Tube A + G*: Treated with **formic acid**, which weakens purine glycosidic bonds (Adenine and Guanine).
  - *Tube C + T*: Treated with **hydrazine**, which cleaves pyrimidine rings (Cytosine and Thymine).
  - *Tube C*: Treated with **hydrazine in high salt (NaCl) solution**, which selectively inhibits Thymine reactions, modifying Cytosine alone.
3. *Backbone Cleavage*:
  - **Piperidine** is added to all four tubes and heated. Piperidine cleaves the phosphodiester backbone specifically at the modified base sites, yielding a collection of labeled fragments of varying lengths.
4. *Electrophoresis and Autoradiography*:
  - The reaction products are loaded into adjacent lanes on a high-resolution polyacrylamide gel (PAGE) containing urea (denaturing conditions) to separate

fragments by size down to single-nucleotide differences.

- An X-ray film is exposed to the gel (autoradiography). The sequence is read from bottom to top (5' to 3' orientation) by noting band occurrences across the four lanes (G, A+G, C+T, C).

**Question** - Write an account of the occurrence and significance of transposable elements in prokaryotes.

**Answer** - Transposable elements (TEs) are specialized mobile DNA segments capable of changing position within a genome. In prokaryotes, these elements occur within genomic DNA, plasmids, and bacteriophages.

#### **Classes of Prokaryotic Transposable Elements:**

1. *Insertion Sequences (IS Elements):*
  - The simplest mobile units (700 to 1,500 base pairs long). They encode only the enzymes required for their own transposition (specifically **Transposase**) and are bounded by inverted terminal repeats (ITRs).
2. *Composite Transposons:*
  - Larger elements flanked by two IS elements that enclose central genes, often carrying resistance markers against antibiotics like tetracycline, chloramphenicol, or kanamycin.
3. *Non-Composite Transposons:*
  - Elements bounded by short inverted repeats that carry structural genes alongside transposase and resolvase enzymes without relying on flanking IS elements.
4. *Transposing Phages (e.g., Bacteriophage Mu):*
  - Temperate phages that integrate into bacterial host chromosomes via replicative transposition, causing mutations upon insertion.

#### **Significance in Prokaryotes:**

1. *Evolution and Dissemination of Antibiotic Resistance:* Transposons drive the rapid accumulation of multi-drug resistance by carrying resistance determinants onto conjugative plasmids (R-factors), accelerating horizontal transfer across bacterial populations.
2. *Mutagenic Impact:* Insertion of a TE into an operon or coding exon disrupts the reading frame, causing knockout mutations or polar effects on downstream genes.
3. *Genomic Rearrangement:* Homologous recombination between multiple copies of identical transposons located across a bacterial chromosome leads to chromosomal inversions, deletions, and duplications.
4. *Gene Regulation Modulation:* Strong promoters embedded within transposable elements can activate nearby silent bacterial genes or alter host transcriptional control.

#### **Question - What are repetitive DNA sequences? How are they differentiated from unique DNA sequences?**

**Answer** - Repetitive DNA sequences are DNA segments present in multiple copies per genome, ranging from two to millions of copies. In contrast, **Unique (Single-Copy) DNA Sequences** occur only once, or a few times, per haploid genome.

#### **Characteristics and Differentiation:**

1. *Copy Number and Genomic Proportion:*
  - *Unique DNA:* Present in low copy number (1 to 2 copies per haploid genome). Constitutes most protein-coding structural genes in eukaryotes.
  - *Repetitive DNA:* Present in hundreds to millions of copies, making up 40-50% or more of complex eukaryotic genomes (e.g., human DNA).
2. *Coding vs. Non-Coding Nature:*

- *Unique DNA*: Encodes vital cellular proteins (e.g., hemoglobin, insulin, enzymes).
- *Repetitive DNA*: Comprises mostly non-coding sequences (satellites, SINEs, LINEs), though it includes multi-copy coding gene families (rRNA, tRNA, Histones).
- 3. *Reassociation Kinetics (C<sub>0</sub>t Curve Behavior)*:
  - When genomic DNA is denatured into single strands and allowed to reanneal:
  - **Repetitive DNA** reassociates **rapidly** because the high concentration of complementary strands makes collision events frequent.
  - **Unique DNA** reassociates **slowly** at higher C<sub>0</sub>t values, requiring more time for complementary strands to find each other.
- 4. *Genomic Density and Location*:
  - *Unique DNA*: Distributed across euchromatic chromosomal arms.
  - *Repetitive DNA*: Highly concentrated in heterochromatic areas such as centromeres and telomeres (satellites), or dispersed throughout intergenic zones (retrotransposons).
- 5. *Rate of Mutation and Polymorphism*:
  - *Unique DNA*: Highly conserved due to evolutionary selection on protein function.
  - *Repetitive DNA*: Exhibits high mutation rates and hypervariability (e.g., VNTRs and STRs), making it useful for genetic profiling.

**Question - Explain the body's defense against invasion with special reference to innate and acquired immunity.**

**Answer** - Immunity is the body's collective defense system for identifying, neutralizing, and removing foreign invaders (pathogens) and abnormal cells. It operates through two integrated branches: **Innate Immunity** and **Acquired (Adaptive) Immunity**.

**1. Innate Immunity (Non-Specific Defense)**: Present from birth, innate immunity provides rapid, immediate protection against a wide range of pathogens without requiring prior exposure.

- *Anatomical/Physical Barriers*: Skin (keratinized layer prevents entry), mucous membranes lining the respiratory, gastrointestinal, and urogenital tracts (mucus traps pathogens).
- *Physiological and Chemical Barriers*: Low gastric pH in stomach juice, lysozyme in tears and saliva, antimicrobial peptides (defensins), and fever, which inhibits microbial replication.
- *Cellular Barriers*: Professional phagocytes (neutrophils, macrophages, monocytes) engulf pathogens using Pattern Recognition Receptors (PRRs) that bind to Pathogen-Associated Molecular Patterns (PAMPs). Natural Killer (NK) cells target virus-infected cells.
- *Cytokine and Complement Systems*: Interferons secreted by virus-infected cells protect neighboring healthy cells. The complement cascade (classical, lectin, and alternative pathways) forms membrane attack complexes (MAC) that lyse bacterial cells.

**2. Acquired Immunity (Specific/Adaptive Defense)**: Acquired immunity develops over time after exposure to specific foreign antigens. It displays high specificity, diversity, self/non-self discrimination, and immunological memory.

- *Humoral Immunity (B-Cell Mediated)*:
  - Driven by B-lymphocytes. Upon antigen recognition via B-cell receptors (BCR), B-cells differentiate into antibody-secreting **plasma cells**.
  - Secreted immunoglobulins (IgG, IgM, IgA, IgE, IgD) neutralize toxins, opsonize bacteria to promote phagocytosis, and activate complement cascades.
- *Cell-Mediated Immunity (T-Cell Mediated)*:
  - Driven by T-lymphocytes. **Helper T-cells (CD4<sup>+</sup>)** process and recognize antigens bound to MHC Class II molecules, secreting cytokines that recruit other immune cells.

- **Cytotoxic T-cells (CD8<sup>+</sup>)** recognize endogenously synthesized antigens bound to MHC Class I molecules, releasing perforins and granzymes to induce apoptosis in infected or malignant cells.
- *Immunological Memory:*
  - Primary exposure generates a reservoir of persistent **Memory B and T cells**. Upon secondary exposure to the same pathogen, these cells mount a faster, stronger, and longer-lasting immune response.

**Question - Explain the mechanism of transfer of genetic material from one bacterial cell to another.**

**Answer -** Bacteria do not undergo true sexual reproduction. Instead, they achieve genetic variation through horizontal gene transfer (HGT), moving genetic material between non-genealogical bacterial cells via three primary mechanisms: **Transformation, Conjugation, and Transduction.**

### 1. Transformation:

- *Mechanism:* The uptake of naked double-stranded DNA fragments directly from the surrounding environment by a competent bacterial cell, followed by integration into the host chromosome through homologous recombination.
- *Process:* Discovered by Frederick Griffith (1928) in *Streptococcus pneumoniae*. Competent cells produce surface protein complexes that bind extracellular DNA, transport a single strand into the cytoplasm, and integrate it into the host genome via RecA-mediated recombination.

### 2. Conjugation:

- *Mechanism:* Direct, contact-dependent transfer of plasmid or genomic DNA from a donor bacterial cell (F<sup>+</sup>) to a recipient cell (F<sup>-</sup>) through a conjugation bridge or sex pilus.
- *Process:*
  - Driven by the **Fertility (F) plasmid**, which encodes sex pili and transfer (tra) genes.
  - The F<sup>+</sup> cell extends a sex pilus to draw the F<sup>-</sup> cell close, forming a cytoplasmic bridge.
  - Relaxosome enzymes cut one DNA strand of the F plasmid at the oriT site. The single strand moves into the recipient cell via rolling circle replication. Both donor and recipient synthesize complementary strands, converting the recipient into an F<sup>+</sup> cell.
  - *High Frequency Recombination (Hfr) Strains:* When the F plasmid integrates into the bacterial chromosome, the host becomes an Hfr strain, capable of transferring chromosomal genes to recipients during conjugation.

### 3. Transduction:

- *Mechanism:* Genetic transfer mediated by a bacteriophage vector.
- *Generalized Transduction:* Occurs during the lytic cycle of virulent phages (e.g., P22, P1). Bacterial chromosomal DNA is accidentally packaged into new phage heads instead of viral DNA. When these phages infect new hosts, they inject bacterial genes.
- *Specialized Transduction:* Occurs during the lysogenic cycle of temperate phages (e.g., Lambda phage). Upon prophage excision from the host genome, adjacent bacterial genes (such as *gal* or *bio*) are excised alongside phage DNA, transferred, and integrated into new recipient genomes.

**Question - What is transgenesis? Describe briefly the technique of gene transfer into the oocyte, egg, and embryo by microinjection.**

**Answer -** Transgenesis is the process of stably introducing an isolated foreign gene (transgene) into the genome of a target organism, giving it new, heritable traits that pass to subsequent

generations.

**DNA Microinjection Technique:** Pronuclear microinjection, developed by Gordon and Ruddle (1980), remains one of the most reliable methods for generating transgenic animals (such as mice, pigs, sheep, and fish).

1. *Collection and Superovulation:*

- Donor female animals undergo hormone treatment (pregnant mare serum gonadotropin followed by human chorionic gonadotropin) to induce superovulation.
- Females are mated with fertile males, and fertilized eggs (zygotes) containing two distinct pronuclei (one male, one female) are harvested from the oviducts.

2. *Microinjection Setup:*

- The harvested zygotes are placed in a specialized depression slide under an inverted light microscope fitted with micromanipulators.
- A fine glass **holding pipette** applies gentle suction to hold the egg firmly in place.
- A fine **injection capillary needle** (tip diameter 0.5 to 1.5 micrometers) loaded with linearized transgene DNA solutions penetrates the cell membrane and enters the **male pronucleus** (which is larger and closer to the surface than the female pronucleus).

3. *DNA Injection:*

- A precise volume (1 to 2 picoliters containing roughly several hundred copies of the construct) is injected into the male pronucleus, causing it to swell slightly.

4. *Embryo Transfer and Development:*

- Injected zygotes are cultured *in vitro* up to the two-cell or blastocyst stage.
- Viable embryos are surgically transferred into the uterus or oviduct of a pseudopregnant surrogate mother (prepared by mating with a vasectomized male).
- The surrogate carries the embryos to term. Transgene integration into host chromosomal DNA occurs randomly during early cleavage stages.

5. *Screening Offspring:*

- Genomic DNA extracted from tail biopsies of newborn pups is screened via PCR or Southern Blotting to confirm successful transgene integration (generating Founder Transgenic Animals).

**Question - Discuss the importance of genetic engineering in medicine.**

**Answer -** Genetic engineering has transformed modern medicine by yielding diagnostic tools, therapeutic proteins, vaccines, and targeted gene therapies.

**Major Applications in Medicine:**

1. *Production of Therapeutic Recombinant Proteins:*

- **Recombinant Human Insulin (Humulin):** Eli Lilly produced human insulin in 1982 by synthesizing A and B chains separately in *E. coli* vectors and joining them via disulfide bonds, replacing slaughterhouse animal insulin that caused allergic reactions.
- **Human Growth Hormone (hGH):** Engineered microbes produce pure hGH to treat pituitary dwarfism, eliminating risks of Creutzfeldt-Jakob disease linked to cadaver-extracted hormone.
- **Clotting Factors and Enzymes:** Recombinant Factor VIII treats Hemophilia A, and Tissue Plasminogen Activator (tPA) dissolves coronary blood clots during acute myocardial infarctions.

2. *Development of Recombinant Vaccines:*

- Genetic engineering produces safer subunit vaccines free of live pathogen risks.

- *Hepatitis B Vaccine*: Produced by expressing Hepatitis B surface antigen (HBsAg) in recombinant transgenic yeast (*Saccharomyces cerevisiae*).
  - *Edible Vaccines*: Genes encoding antigenic proteins inserted into banana or potato plants induce mucosal immunity upon consumption.
3. **Gene Therapy**:
- Directly corrects underlying genetic disorders by inserting functional copies of missing or mutated genes.
  - *Severe Combined Immunodeficiency (SCID)*: Successfully treated using retroviral vectors to deliver functional Adenosine Deaminase (ADA) genes into patients' autologous hematopoietic T-lymphocytes.
4. **Monoclonal Antibodies and Molecular Diagnostics**:
- Recombinant monoclonal antibodies (e.g., Trastuzumab for HER2<sup>+</sup> breast cancer, Rituximab for lymphoma) target cancer cells with high specificity.
  - PCR and real-time RT-PCR allow rapid, accurate detection of low-copy infectious pathogens (e.g., HIV, SARS-CoV-2) and inherited genetic mutations.

**Question - Discuss the 'chain termination' method of DNA sequencing. How does it differ from the enzymatic degradation method?**

**Answer** - The "Chain Termination Method," developed by Nobel Laureate Frederick Sanger in 1977, is an enzymatic approach for determining DNA sequences. It remains the foundation of automated capillary sequencing technology.

**Mechanism of Chain Termination Method:**

1. **Reaction Components**: Single-stranded DNA template, a complementary synthetic primer, DNA Polymerase enzyme, standard Deoxynucleotide Triphosphates (dATP, dCTP, dGTP, dTTP), and small amounts of modified **Dideoxynucleotide Triphosphates** (ddATP, ddCTP, ddGTP, ddTTP).
2. **Role of Dideoxynucleotides (ddNTPs)**:
  - ddNTPs lack the critical 3'-OH group on their ribose sugar ring, featuring a 3'-H instead.
  - When DNA Polymerase incorporates a ddNTP into a growing DNA strand, no 3'-OH group is available to form the next 5' to 3' phosphodiester bond, **terminating chain elongation immediately**.
3. **Process**: Four separate reaction mixtures are prepared, each containing all four dNTPs and one specific ddNTP. Over time, random incorporation of ddNTPs yields a population of variable-length DNA fragments, each ending at that specific nucleotide base.
4. **Resolution**: Reaction products are separated by size on a high-resolution polyacrylamide gel, and the complementary sequence is read from bottom (5') to top (3').

**Comparison with Chemical Degradation Method (Maxam-Gilbert Method):**

- **Reaction Mechanism**: Sanger's method relies on **enzymatic synthesis** and selective chain termination. Maxam-Gilbert sequencing relies on base-specific **chemical cleavage** of pre-existing, radiolabeled DNA chains.
- **Reagent Hazards**: Sanger's method uses non-hazardous enzymes and nucleotides. Maxam-Gilbert requires toxic compounds like dimethyl sulfate and hydrazine.
- **Automation and Scalability**: Sanger's method adapts easily to fluorescent dye-labeling and automated capillary electrophoresis (yielding modern ABI sequencers). Chemical cleavage is labor-intensive and unsuitable for high-throughput sequencing.

**Question - Describe the regulation of lac operon. And explain what is catabolite repression.**

**Answer** - The *Lac Operon* of *E. coli*, described by François Jacob and Jacques Monod in 1961,

serves as the classic model for inducible gene regulation in prokaryotes. It contains structural genes encoding enzymes required for lactose transport and breakdown.

### Structural Components of Lac Operon:

1. *Structural Genes:*
  - *lacZ*: Encodes  $\beta$ -galactosidase (cleaves lactose into glucose and galactose).
  - *lacY*: Encodes  $\beta$ -galactoside permease (membrane protein that transports lactose into the cell).
  - *lacA*: Encodes  $\beta$ -galactoside transacetylase (transfers an acetyl group to  $\beta$ -galactosides).
2. *Regulatory Sequence:* Promoter (P), Operator (O), and an independent regulatory gene (*lacI*) encoding the **Lac Repressor protein**.

### Regulation Mechanism:

1. *Negative Control (Absence of Lactose):*
  - The *lacI* gene continuously produces active tetrameric Lac Repressor protein.
  - The repressor binds tightly to the operator (O) sequence, physically blocking RNA Polymerase from transcribing structural genes (*lacZYA*). The operon stays **OFF**.
2. *Induction (Presence of Lactose / Allolactose):*
  - Lactose entering the cell is converted into **allolactose** (the true inducer).
  - Allolactose binds to the Lac Repressor, inducing a conformational change that prevents it from binding to the operator.
  - RNA Polymerase binds the promoter and transcribes *lacZYA* mRNA. The operon turns **ON**.

**Catabolite Repression (Positive Control by Glucose):** Glucose is *E. coli*'s preferred energy source. When glucose and lactose are both present, bacteria selectively utilize glucose first, suppressing *lac* operon expression through **catabolite repression**.

- **High Glucose Levels:** Adenylate Cyclase activity is inhibited, resulting in **low cyclic AMP (cAMP)** levels. Without cAMP, the **Catabolite Activator Protein (CAP)** remains inactive and cannot bind the CAP site near the promoter. RNA Polymerase binds weakly, keeping *lac* transcription minimal.
- **Low Glucose / High Lactose Levels:** Adenylate Cyclase activates, raising **cAMP levels**. cAMP binds CAP to form an active cAMP-CAP complex, which attaches to the CAP binding site. This bends the DNA, helping RNA Polymerase bind the promoter and triggering robust *lac* operon transcription.

**Question - What are split genes? How are these visualised in an eukaryotic genome?**

**Answer - A Split Gene** is a gene whose coding nucleotide sequence is interrupted by non-coding sequences. Discovered in 1977 by Richard Roberts and Phillip Sharp (using adenovirus models), split genes occur across eukaryotic genomes.

### Structural Organization:

- **Exons:** Expressed sequences that remain in mature mRNA and code for amino acids in the final protein.
- **Introns:** Intervening non-coding sequences transcribed into pre-mRNA but spliced out before protein synthesis.

### Visualization of Split Genes in Eukaryotic Genomes:

1. *R-Looping Technique (Heteroduplex Mapping under Electron Microscopy):*
  - Historically, split genes were visualized by hybridizing mature mRNA with its corresponding single-stranded genomic DNA template.
  - The coding exons of genomic DNA hybridize continuously with mature mRNA.
  - The non-coding **introns** cannot pair with mature mRNA (since introns were spliced

out). As a result, intron regions loop outward as single-stranded DNA structures known as **R-loops**.

- Transmission electron microscopy reveals these loops; the number of single-stranded R-loops corresponds directly to the number of introns present in the gene.
- 2. *Gel Electrophoresis and Northern/Southern Blot Comparison:*
  - Comparing cDNA lengths (synthesized from spliced mRNA via reverse transcriptase) against genomic DNA fragments via gel electrophoresis reveals that genomic DNA is substantially longer than mature mRNA due to intron presence.
- 3. *In Silico Bioinformatic Visualization:*
  - Modern computational tools match genomic sequences against cDNA or expressed sequence tags (ESTs) using alignment algorithms (like BLAST). Intron-exon junction splice sites (5'GU...AG3') appear mapped visually as spliced gaps across genomic alignment tracks.

**Question - Describe the application of recombinant DNA technology in agriculture.**

**Answer -** Recombinant DNA (rDNA) technology has transformed modern agriculture by enabling the targeted transfer of beneficial traits across species boundaries, yielding Genetically Modified (GM) crops with enhanced pest resistance, stress tolerance, and nutritional content.

**Major Applications:**

1. *Insect and Pest Resistance:*
  - **Bt Crops:** Transgenic plants incorporating cry genes from *Bacillus thuringiensis*. When insect larvae ingest crop tissue,  $\delta$ -endotoxin proteins solubilize in their alkaline midgut, binding receptors and forming lethal pores. This confers endogenous resistance against pink bollworm in cotton (Bt Cotton) and fruit borers in brinjal.
2. *Herbicide Tolerance:*
  - Crop plants engineered with genes enabling survival after broad-spectrum herbicide applications. For example, "Roundup Ready" crops incorporate a bacterial gene (EPSPS synthase from *Agrobacterium*) resistant to glyphosate, allowing effective weed control without crop damage.
3. *Nutritional Fortification (Biofortification):*
  - **Golden Rice:** Engineered with three biosynthetic pathway genes (two from daffodil, one from *Erwinia*) to synthesize  $\beta$ -carotene in grain endosperm, helping combat Vitamin A deficiency in developing nations.
  - **Protein Enrichment:** Transgenic potatoes incorporating the AmA1 gene from *Amaranthus hypochondriacus* produce significantly higher levels of essential amino acids.
4. *Abiotic Stress Tolerance:*
  - Transferring genes regulating osmoprotectants (like proline and glycine betaine) or heat shock proteins helps crops tolerate drought, high salinity, and extreme temperatures.
5. *Delayed Fruit Ripening:*
  - Flavr Savr tomatoes were engineered with antisense RNA technology to silence the polygalacturonase enzyme gene, which breaks down cell wall pectin. This delayed softening, extending shelf life and preserving flavor during transport.

**Question - Describe the Maxam Gilbert method of DNA sequencing and compare it with Sanger's method.**

**Answer -** The Maxam-Gilbert method (1977), or chemical cleavage method, determines

nucleotide sequences by using base-specific chemical modifications and phosphodiester backbone cleavage.

#### Maxam-Gilbert Method Overview:

- 5' end-labeled single-stranded DNA is divided into four aliquots (G, A+G, C+T, C).
- Base-modifying chemicals (dimethyl sulfate for G, formic acid for A+G, hydrazine for C+T, and hydrazine with salt for C) are added at low concentrations to modify approximately one target base per molecule.
- **Piperidine** treatment cleaves the phosphodiester backbone at modified sites.
- Gel electrophoresis (PAGE) and autoradiography resolve fragments down to single-base differences, and the sequence is read directly from the band patterns across lanes.

#### Comparison Between Maxam-Gilbert and Sanger Methods:

1. *Basic Reaction Principle:* Maxam-Gilbert uses **selective chemical degradation** of pre-existing DNA strands. Sanger uses **selective enzymatic synthesis** paired with chain-terminating dideoxynucleotides (ddNTPs).
2. *Reagent Safety:* Maxam-Gilbert relies on toxic, hazardous compounds (dimethyl sulfate, hydrazine, piperidine). Sanger uses benign enzymes and nucleotide analogs.
3. *DNA Template Requirement:* Maxam-Gilbert works with double-stranded or single-stranded DNA but requires clean end-labeling  $^{32}\text{P}$ . Sanger requires single-stranded DNA templates (or denatured double-stranded DNA) and a synthetic primer.
4. *Read Length and Accuracy:* Maxam-Gilbert reads are typically shorter and prone to incomplete cleavage artifacts. Sanger provides longer, cleaner sequencing reads.
5. *Automation Potential:* Maxam-Gilbert cannot be easily automated. Sanger sequencing adapted seamlessly to fluorescent dye labeling and capillary gel systems, driving high-throughput automated sequencing platforms (like the Human Genome Project).

#### Question - What is chromosome mapping? Describe the method of chromosome mapping in bacteria.

**Answer** - Chromosome mapping is the process of determining the relative locations of genes on a chromosome and measuring the linear distances between them. Distances are derived from recombination frequencies or conjugation transfer times.

#### Methods of Chromosome Mapping in Bacteria:

Bacteria lack standard meiotic crossing-over. Instead, mapping relies on three primary mechanisms:

**1. Interrupted Mating Conjugation (Hfr  $\times$  F<sup>-</sup> Mapping):** Discovered by Jacob and Wollman, this approach measures gene transfer time during bacterial conjugation.

- *Procedure:* An Hfr donor strain containing wild-type dominant alleles (str<sup>S</sup> a<sup>+</sup> b<sup>+</sup> c<sup>+</sup> d<sup>+</sup>) is mixed with an F<sup>-</sup> recipient strain carrying recessive mutant markers (str<sup>R</sup> a<sup>-</sup> b<sup>-</sup> c<sup>-</sup> d<sup>-</sup>).
- Conjugation begins as the Hfr chromosome transfers sequentially into recipients at a constant speed, starting from the oriT site.
- Samples removed at 1-minute intervals are placed in a blender to shear sex pili and interrupt transfer.
- Samples plated on streptomycin selective agar kill Hfr donor cells while permitting recombinant recipient survival.
- *Mapping Metric:* Plotting the arrival time of each gene yields a map measured in **minutes** (the *E. coli* chromosome corresponds to a 100-minute circular map).

#### 2. Generalized Transduction Mapping:

- Bacteriophages (e.g., P1) occasionally package host chromosomal fragments instead of viral genomes.

- If two genes sit close together on a chromosome, they are likely to be packaged into a single phage head (**co-transduction**).
- Measuring co-transduction frequencies reveals fine-structure genetic distances: higher co-transduction frequencies reflect closer physical proximity.

### 3. Transformation Mapping:

- Similar to transduction mapping, the frequency with which two donor markers undergo joint transformation (**co-transformation**) into a competent host cell provides fine-scale map distances for closely linked genes.

**Question - What do you mean by cloning vector? What are the different types of vectors used in gene cloning? Give a detailed description of any one of them.**

**Answer - A Cloning Vector** is a small, stably maintained DNA molecule used as a vehicle to carry foreign genetic material into a host cell, where it can be replicated and expressed independently.

#### Essential Requirements of a Cloning Vector:

1. An **Origin of Replication (ori)** allowing autonomous replication inside host cells.
2. A **Selectable Marker Gene** (e.g., antibiotic resistance) to distinguish transformed host cells from non-transformed cells.
3. Unique **Restriction Enzyme Sites (Multiple Cloning Site - MCS)** for inserting target genes.

#### Types of Cloning Vectors:

- **Plasmids:** Small, circular double-stranded extra-chromosomal DNA (capacity up to 10 kb; e.g., pBR322, pUC19).
- **Bacteriophage Vectors:** Derived from viruses like Lambda ( $\lambda$ ) phage (capacity 10–20 kb).
- **Cosmids:** Hybrid vectors combining plasmid sequences with phage  $\lambda$  cos sites (capacity 30–45 kb).
- **Bacterial Artificial Chromosomes (BACs):** Based on the F-factor plasmid (capacity 100–300 kb).
- **Yeast Artificial Chromosomes (YACs):** Linear vectors containing yeast centromeric, telomeric, and ARS sequences (capacity 100–1000 kb).

**Detailed Description of Plasmid Vector pBR322:** Constructed by Francisco Bolivar and Raymond Rodriguez in 1977, **pBR322** was one of the earliest purpose-built multipurpose plasmid vectors.

- **Size:** 4,361 base pairs long.
- **Origin of Replication (ori):** Derived from plasmid pMB1, ensuring relaxed control with high copy numbers (15–20 copies per cell).
- **Selectable Markers:**
  - Contains two distinct antibiotic resistance genes: the **Ampicillin resistance gene (amp<sup>R</sup>)** and the **Tetracycline resistance gene (tet<sup>R</sup>)**.
- **Unique Restriction Sites:**
  - Contains distinct recognition sites for restriction endonucleases.
  - Inserting a foreign DNA fragment into the *Bam*HI or *Sal*I site located within the tet<sup>R</sup> gene inactivates tetracycline resistance (**Insertional Inactivation**).
  - Transformants carrying recombinant plasmids remain resistant to ampicillin (amp<sup>R</sup>) but become sensitive to tetracycline (tet<sup>S</sup>), making them easy to identify by replica plating.